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评述

开放环境下的协作多智能体强化学习进展

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摘要 多智能体强化学习 (multi-agent reinforcement learning, MARL) 近年来获得广泛关注并在不同领域取得进展. 其中, 协作多智能体强化学习专注于训练智能体团队以协同完成单智能体难以应对的任务目标, 在路径规划、无人驾驶、主动电压控制和动态算法配置等场景展现出巨大的应用潜力. 如何提升系统协作效能是协作多智能体强化学习领域研究重点之一. 以往的研究工作主要在简单、静态和封闭的环境设定中展开. 随着人工智能技术落地的驱使, 目前在多智能体协作领域也有部分研究开始对开放环境下的多智能体协作展开研究, 这些工作从多个方面对智能体所处环境中要素可能发生改变这一情况进行探索与研究, 并取得一定进展. 但是当前主流工作仍然缺乏对该方向的综述. 本文从强化学习概念着手, 针对多智能体系统、协作多智能体强化学习、典型方法与测试环境进行介绍, 对封闭到开放环境下的协作多智能体强化学习研究工作进行总结, 提炼出多类研究方向并对典型工作进行介绍. 最后, 本文对当前研究的优势与不足进行了总结, 对未来开放环境下协作多智能体强化学习的发展方向与待研究问题进行展望, 以吸引更多研究人士参与这个新兴方向的研究与交流.

关键词 强化学习, 多智能体系统, 多智能体协作, 开放环境机器学习, 开放环境多智能体协作

1 引言

作为机器学习的一个分支, 强化学习 (reinforcement learning, RL) [1] 是一种解决序贯决策问题的有效方法, 相较于监督学习与无监督学习, 其显著的特点在于从交互中进行学习. 在强化学习范式中, 智能体通过与环境进行交互, 根据所获得的奖赏或惩罚不断优化其策略. 由于其学习方式与人类获取知识的方式类似, 强化学习被视为实现通用人工智能 (artificial general intelligence, AGI) [2] 的重要途径之一. 早期的强化学习工作依赖于手工特征输入到线性模型进行估值和拟合, 在复杂场景中表现不佳. 在过去的十年中, 得益于深度学习 [3] 的蓬勃发展, 深度强化学习在各行各业取得瞩目成就. 例如, 深度 Q- 网络 (deep Q-network, DQN) [4] 在雅达利 (Atari) 视频游戏上超越了人类职业玩家; AlphaGo [5] 在围棋人机大战中击败围棋世界冠军李世石; AlphaStar [6] 在不完全信息即时战略游戏星际争霸 II 中战

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胜了人类顶尖职业选手; OpenAI Five [7] 在多人实时在线竞技游戏 Dota 2 中表现出色; 而 AI-Suphx [8] 在多人非完全信息博弈麻将中也取得了重要的成果. 此外, 强化学习的应用范围也逐渐从游戏领域扩展到现实生活的各个领域, 包括工业制造、机器人控制、物流管理、国防军事、智慧交通、智能医疗等, 极大地推动了人工智能技术的发展 [9,10], 例如, 最近得到广泛关注的 ChatGPT [11] 也使用了强化学习技术进行优化. 近年来, 在人工智能应用于科学研究 (AI4Science) [12] 的发展趋势下, 强化学习在许多基础科学领域也大放异彩, 如 DeepMind 利用深度强化学习实现了核聚变的控制 [13], 发现矩阵乘法的强化学习方法 AlphaTensor [14] 等.

与此同时, 现实中的很多问题往往是大规模、复杂、实时并带有不确定性的, 将此类问题建模为单智能体问题效率低下且与现实条件不符, 而将其建模为多智能体系统 (multi-agent system, MAS) [15] 问题则往往更加适配. 进一步, 由于很多复杂问题难以由某个个体独立完成, 需要多智能体协同, 如自动驾驶汽车、智能仓储系统和传感器网络等. 协作多智能体强化学习 (multi-agent reinforcement learning, MARL) [16~18] 为这些问题的建模和解决提供了强有力的支撑. 在这类问题中, 智能体团队通过与环境交互学习一个联合的协作策略以解决任务. 相较于传统算法, 多智能体强化学习的优势在于能够应对环境的不确定性, 并在不需要过多领域知识的条件下学习解决未知任务. 近年来, 深度学习与多智能体强化学习的结合产出了丰硕的成果 [19], 许多算法被提出并用于解决复杂任务. 然而, 多智能体强化学习区别于单智能体问题的特殊性也带来了新的挑战. 一方面, 真实多智能体系统所处环境往往是部分可观测的, 单个智能体无法从其局部观测中获得环境的全局信息, 这意味着独立学习的智能体难以做出最优决策 [20]. 另一方面, 由于其他智能体同时进行学习, 策略相应地会发生变化, 从单个智能体角度来看, 其处于一个非稳态的环境中, 收敛性无法得到保证 [21]. 此外, 协作型多智能体系统往往只能得到共享奖赏, 如何将其分配从而为每个智能体提供准确的反馈, 进而使其高效地学习协作, 最终让系统性能得到最大提升也是目前存在的核心挑战之一 [22]. 最后, 随着多智能体系统的扩充, 智能体的数量不断增加, 解决强化学习问题所面临的搜索空间将呈指数增长, 这使得在策略空间中搜索和学习变得极为困难. 因此, 如何组织高效的策略学习, 也是目前的一大难点 [23,24].

针对前述挑战, 目前的大量工作从多方面展开研究, 并在许多任务场景中取得令人惊喜的成果. 在路径规划 [25]、主动电压控制 [26] 和动态算法配置 [27] 等任务中, 协作型多智能体强化学习都展现出比传统方法更优秀的性能. 研究者设计出许多方法以促进智能体之间的协作, 包括基于策略梯度的方法, 如 MADDPG [28] 和 MAPPO [29]; 基于值函数的方法, 如 VDN [30] 和 QMIX [31]; 或包括借助 Transformer 的强大表达能力提升协作能力的 MAT [32] 在内的其他方法. 这些方法已经在许多任务 (如 SMAC [33], Hanabi 和 GRF [29]) 中表现出优秀的协作能力. 除了上述方法和相应的变体外, 研究者亦从其他角度对协作型 MARL 进行了深入探索与研究, 包括通过高效通信 [20] 缓解分布式策略执行设定下的部分可观测性、对策略进行离线部署 [34]、MARL 中的世界模型学习 [35] 和训练范式研究 [36] 等.

传统的机器学习研究通常在经典封闭环境情景这一假设下进行, 其中学习过程的重要因素保持不变. 如今越来越多的实际任务, 尤其是涉及开放环境情景的任务, 其中的重要学习因素可能会发生变化. 显然, 对于机器学习来说, 从经典环境转向开放环境是一个巨大的挑战. 针对数据驱动的学习任务, 开放环境下数据会随着时间在线地积累, 例如, 数据流这一形式, 使得模型的学习更具挑战性. 开放环境下的机器学习 [37,38] 在许多场景具有应用前景, 逐渐引起广泛关注. 在机器学习领域, 目前的研究工作主要包括类别变化、特征增减、数据分布变化、学习目标变化等. 与之对应, 强化学习领域的部分工作也开始着眼于开放环境下的任务, 主要工作包括可信强化学习 [39]、环境生成与策略学习 [40]、持续强化学习 [41]、强化学习泛化能力 [42]、元强化学习 [43] 与模拟器到真实环境的策略迁移 [44] 等.

相较于单智能体强化学习, 多智能体场景更复杂也更有挑战性, 目前对开放环境下的协作多智能体研究较少, 有部分工作开始关注多智能体的稳健性研究 [45], 并从不同角度进行问题描述与算法设计 [46~49]. 此外, 针对开放团队的多智能体强化学习问题, 一些工作提出了包括即时团队协作 (ad-hoc teamwork, AHT) [50]、零样本协作 (zero-shot coordination, ZSC) [51] 和少样本团队协作 (few-shot

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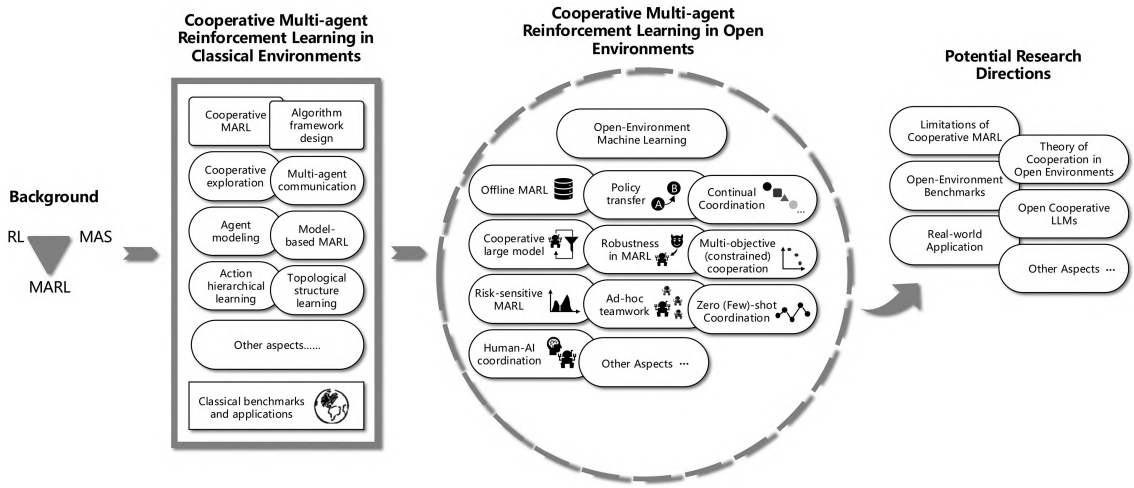

图 1　本文整体架构.

Figure 1　Overall structure of the article.

teamwork, FST) [52] 在内的设定, 以应对这一挑战. 虽然上述工作在一些任务场景取得了成效, 然而其与大部分真实应用场景仍然不相匹配并且实际效果存在较大提升空间. 目前针对多智能体强化学习已经存在部分综述性工作, 比如多智能体系统 [15]、多智能体强化学习 [16,53~58]、多智能体建模 [59]、多智能体中的非稳态处理 [21]、多智能体中的迁移学习 [60]、协作多智能体强化学习 [17,61,62]、多智能体模型学习 [35]、因果多智能体强化学习 [63]、多智能体通信 [20] 等. 此外, 部分工作也对开放机器学习进行了综合分析 [37,38,64]. 虽然上述工作从多智能体强化学习亦或是开放环境机器学习方面都进行了一些回顾和归纳, 但是针对开放环境下的多智能体强化学习, 尤其开放环境下的协作多智能体强化学习, 目前没有工作进行系统性综述. 考虑到协作多智能体强化学习在解决真实环境中复杂协作问题的潜力与价值, 本文拟对最近的工作进展进行描述. 本文后续的安排如图 1 所示, 首先介绍本文涉及的背景知识, 包括强化学习的基本知识、多智能体系统到多智能体强化学习常见知识与背景; 之后, 对经典环境下的协作多智能体强化学习进行介绍, 从具体定义到当前主流研究内容, 以及常见测试环境与应用案例; 接下来, 对开放环境下的协作多智能体强化学习进行介绍, 具体而言, 包括封闭机器学习、强化学习到协作多智能体常见研究方向与内容; 最后总结本文的主要内容, 并对开放环境下的协作多智能体协作进行展望, 以期为该方向研究思路与方法的拓展提供抛砖引玉的作用, 引起更多研究者对开放环境下的协作多智能体强化学习的关注.

2　背景知识

2.1　强化学习

强化学习 [1] 旨在指导智能体依据当前的状态学习执行恰当的动作, 即学习到一个观测状态到动作的映射, 通过决策以最大化环境反馈的累积数值奖赏 (reward). 环境会依据当前状态和智能体执行的动作反馈其相应的奖赏信息. 智能体无法得知要执行的最优动作, 而必须通过尝试从而发现那些能产生最大累积奖赏的动作. 在标准 RL 场景中, 智能体通过观测状态和执行动作与环境交互. 在交互的每个时间步, 智能体都会接收对环境当前状态的观测, 然后依据该观测选择动作作为输出. 上述动作的执行会改变环境的状态, 并发送给智能体环境反馈的奖赏信号. 智能体的目的是执行能最大化累积数值奖赏的动作序列.

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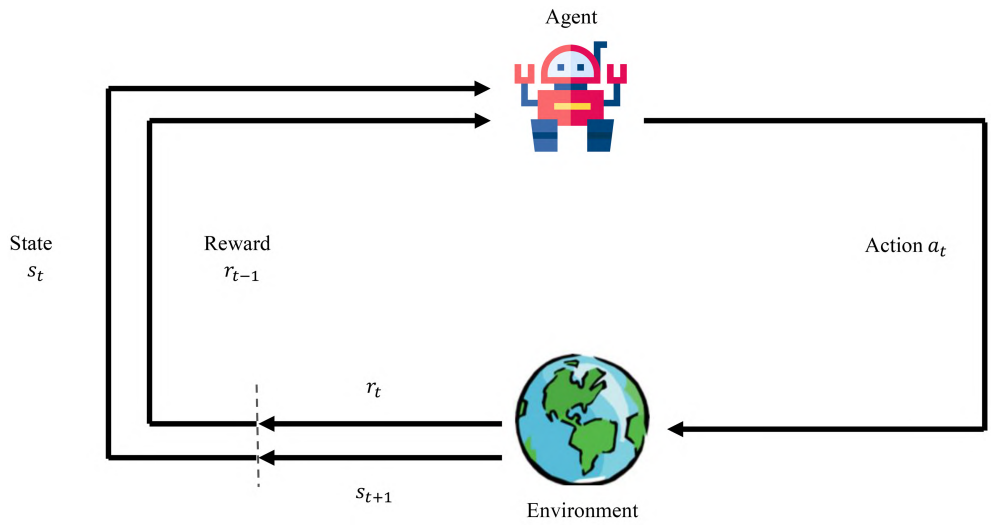

图 2　(网络版彩图) 强化学习示意图.

Figure 2　(Color online) Diagram of reinforcement learning.

2.1.1　问题设定

强化学习 [1] 是机器学习的一个分支, 相较于机器学习中经典的监督学习和无监督学习, 其最大特点是在交互中学习, 即智能体在与环境的交互中根据获得的奖赏或惩罚不断优化策略进而适应新的环境. 强化学习主要由 4 个部分构成: 智能体 (agent)、状态 (state)、动作 (action) 和奖赏 (reward) (如图 2 所示). 强化学习的目标是最大化累积奖赏, 智能体需要通过反复探索试错并学习, 找到对应的最优策略, 该过程一般可以建模为马尔可夫决策过程 (Markov decision process, MDP).

定义1 (马尔可夫决策过程)　马尔可夫决策过程由五元组 $\langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$ (无限过程) 或五元组 $\langle \mathcal{S}, \mathcal{A}, P, R, T \rangle$ (有限过程) 组成, 其中

- $\mathcal{S}$ 为所有状态的集合,
- $\mathcal{A}$ 为所有动作的集合,
- $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ 为状态转移概率函数: $P(s' \mid s, a) = \Pr[S_{t+1} = s' \mid S_t = s, A_t = a]$,
- $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ 为奖赏函数: $R(s, a) = \mathbb{E}[r_t \mid S_t = s, A_t = a]$,
- $\gamma \in [0, 1]$ 是折扣因子, T 是最大轨迹长度.

智能体可以自由地选择在一个状态上执行哪个动作, 或者按照一定的概率分布来选择. 这里强调 "自由选择", 即是智能体的 "能动性", 可以自主地改变动作的选择. 对于一种固定下来的动作选择, 本文称之为策略.

具体而言, 在任一时间步 t, 智能体从环境中观测到当前状态 s_t, 然后执行动作 a_t. 该动作使得智能体所处的环境依据转移函数 $s_{t+1} \sim P(\cdot|s_t, a_t)$ 转移至新的状态, 并且接收环境反馈的奖赏信息 $r_t = R(s_t, a_t)$. 不失一般性, 本文主要讨论无限过程的 MDP, 智能体的目标是: 找到最优策略以最大化累积奖赏. 在数学上, 上述过程可以总结为让智能体找到一个马尔可夫的 (输出仅取决于当前状态) 且平稳的 (函数形式与时间无关) 策略函数 $\pi : \mathcal{S} \to \Delta(\mathcal{A})$, 从而引导智能体执行恰当的序列决策, 最大化累积奖赏, 其中 $\Delta(\mathcal{A})$ 是动作空间上的概率分布, 优化目标, 即

$$\mathbb{E}^{\pi}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0\right], \tag{1}$$

其中 $\mathbb{E}^{\pi}[\cdot]$ 是针对策略 $\pi(a_t|s_t)$ 和状态转移 $P(s_{t+1}|s_t, a_t)$ 下得到的序列 $\tau = (s_0, a_0, s_1, a_1, \ldots)$ 的分布计算所得的期望.

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依据式 (1) 的目标函数, 在策略 π 的指导下, 可以定义状态 – 动作价值函数, 即 Q- 函数. 它表示在策略 π 下, 基于状态 s 执行动作 a 后的期望累积奖赏; 此外, 状态值函数 V 描述的是基于状态 s 的期望累积奖赏:

$$\begin{aligned} Q^{\pi}(s,a) &= \mathbb{E}^{\pi}\left[\sum_{t=0}^{\infty}\gamma^{t}R(s_t,a_t) \mid s_0=s, a_0=a\right], \\ V^{\pi}(s) &= \mathbb{E}^{\pi}\left[\sum_{t=0}^{\infty}\gamma^{t}R(s_t,a_t) \mid s_0=s\right]. \end{aligned} \tag{2}$$

显然, 状态动作函数 Q- 函数和状态价值函数 V- 函数之间的关系可表示为 $V^{\pi}(s)=\mathbb{E}_{a\sim\pi(\cdot|s)}\left[Q^{\pi}(s,a)\right]$ 和 $Q^{\pi}(s,a)=\mathbb{E}_{s'\sim P(\cdot|s,a)}\left[R(s,a)+V^{\pi}(s')\right]$. 基于两类值函数的定义, 可以将强化学习在马尔可夫决策过程中的学习目标表示为寻找最优的策略 π_* 以最大化价值函数: $\pi_*=\arg\max_{\pi}V^{\pi}(s),\forall s$.

2.1.2　基于值函数的强化学习

对于状态空间和动作空间有限的马尔可夫决策过程, 至少存在一个确定性平稳最优策略使累积奖赏最大化. 而基于值函数的算法需要构建并估计值函数, 该值函数随后用于指导动作的选择和执行, 导出对应的策略. 此类算法大多都不直接使用 V- 函数, 而需使用 Q- 函数, 使得式 (2) 对应的 Q- 值最大化. 相应地, 通过取贪心动作, 由 Q- 函数诱导出的最优策略可以表示为 $\pi^*(a|s)=\mathbb{1}\{a=\arg\max_a Q^*(s,a)\}$, 其中 $\mathbb{1}\{\cdot\}$ 为指示函数. 经典的 Q- 学习算法通过时序/时间差分 (temporal-difference, TD) 学习代理函数 $\hat{Q}$- 值 [65], 以近似最优函数 Q^*, 更新过程为

$$\hat{Q}(s_t,a_t)\leftarrow\hat{Q}(s_t,a_t)+\alpha\cdot\overbrace{\left(\underbrace{r_t+\gamma\max_{a_{t+1}\in\mathcal{A}}\hat{Q}(s_{t+1},a_{t+1})}_{\text{差分目标}}-\hat{Q}(s_t,a_t)\right)}^{\text{差分误差}}. \tag{3}$$

理论上, 给定贝尔曼 (Bellman) 最优算子 $\mathcal{B}^*$ 后, Q- 函数的求解过程可以被定义为 [66]

$$(\mathcal{B}^*Q)(s,a)=\sum_{s'}P(s'|s,a)\left[R(s,a)+\gamma\max_{a'\in\mathcal{A}}Q(s',a')\right]. \tag{4}$$

为了建立最优的 Q- 函数, Q- 学习算法利用贝尔曼方程的不动点迭代来求解具有 (唯一) 解的 Q- 函数 $Q^*(s,a)=(\mathcal{B}^*Q^*)(s,a)$. 在实践中, 当环境模型不可知, 状态 – 动作对均为离散表示, 且所有动作在所有状态下可重复采样的情况下, 上述 Q- 学习方法可以保证收敛到最优解.

由于许多现实问题的环境可能具有连续的、高维的状态 – 动作空间, 导致上述假设通常不适用. 此时, 智能体需要对参数化的 Q- 函数 $Q(s,a|\theta)$ 进行学习, 其中 θ 是函数实例化的参数. 为了更新 Q- 函数, 智能体需要以元组 (s,a,r,s') 的形式收集其与环境交互时产生的样本序列. 其中奖赏 r 和下一个时间步的状态 s' 遵循环境基于状态 – 动作对 (s,a) 的反馈. 每次迭代时, 当前的 Q- 函数 $Q(s,a|\theta)$ 可以进行如下更新 [67]:

$$\begin{aligned} y^Q &= r+\gamma\max_{a'\in\mathcal{A}}Q(s',a'|\theta), \\ \theta &\leftarrow \theta+\alpha\left(y^Q-Q(s,a|\theta)\right)\nabla_{\theta}Q(s,a|\theta), \end{aligned} \tag{5}$$

其中 α 是更新速率.

式 (5) 中的 Q- 学习方式可以直接使用神经网络 $Q(s,a|\theta)$ 拟合, 以向最优 Q^*- 值收敛, 其中参数 θ 通过随机梯度下降 (或其他优化方式) 更新. 然而, 由于神经网络的泛化和推理能力有限, Q- 网络在状态 – 动作空间的不同位置会产生不可预测的变化. 因此, 式 (4) 中的贝尔曼算子的收缩映射性质不

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足以保证收敛性. 大量实验证明, 这些误差会随着在线更新规则的传播而传导, 从而导致收敛速度缓慢甚至不稳定. 使用函数逼近 Q- 值的另一不利影响是, 由于 max 算子的作用, Q- 值往往会被过高估计. 因此, 由于不稳定性和过高估计的风险, 必须特别注意保证适当的学习速率. Mnih 等 [68] 提出的 DQN 学习算法使用了两个重要元素以缓解上述不利影响: 目标 Q- 网络和经验回放/重放. 具体来说, DQN 实例化一个参数为 θ^- 的目标 Q- 网络, 用以计算差分目标; 同时利用经验回放池 $\mathcal{D}$ 存储采样收集的样本序列, 不仅保证了样本利用率, 同时在训练时通过独立同分布的采样以缓解采样数据的相关性. DQN 的优化目标由此可以表示为

$$\min_{\theta} \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\left(r+\gamma\max_{a'\in\mathcal{A}} Q(s',a'|\theta^-)-Q(s,a|\theta)\right)^2\right], \tag{6}$$

其中目标 Q- 网络 $Q(s,a|\theta^-)$ 会定期地更新以同步于 Q- 网络的参数. 通过上述方式以及两个重要元素可以实现对 Q- 网络的稳定训练.

此外, 近期有许多工作对 DQN 进行了额外的改进: 文献 [69] 利用神经网络结构解耦了值函数和优势函数 (定义为 $A(s,a)=Q(s,a)-V(s)$), 从而提高了学习性能. 在文献 [70] 中, 一种特定的更新方案 (式 (6) 的变体) 可以减少对 Q- 值的高估, 同时提高学习的稳定性. 并行学习 [71] 或使用无监督辅助任务 [72] 也有助于更快、更稳健地学习. 在文献 [73] 中, 可微记忆模块可以通过在蒙特卡洛值估计和异策略估计之间进行插值, 从而快速集成最近的经验.

2.1.3　基于策略梯度的强化学习

基于策略的方法 (policy-based methods) [74] 的基本思想是直接学习参数化的最优策略 π_θ. 与基于值函数的方法相比, 基于策略的算法具有以下特性: 首先, 这类方法可以灵活地应用于连续动作空间中; 另外, 此类方法可以直接获得随机策略 $\pi_\theta(\cdot|s)$. 当用神经网络参数化策略时, 典型方法是朝着提高累积奖赏的方向调整参数: $\theta\leftarrow\theta+\alpha\nabla_\theta V^{\pi_\theta}(s)$. 然而, 梯度也将受制于策略变化对状态分布的未知影响. 早在文献 [1] 中, 研究者基于策略梯度 (policy gradient) 定理推导出了一个不涉及状态分布的解:

$$\nabla_\theta V^{\pi_\theta}(s)=\mathbb{E}_{s\sim\mu^{\pi_\theta}(\cdot),a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta\log\pi_\theta(a|s)\cdot Q^{\pi_\theta}(s,a)\right], \tag{7}$$

其中 μ^{π_θ} 是在策略 π_θ 下状态的占用度量 [75], $\nabla_\theta\log\pi_\theta(a|s)$ 是策略的更新得分评估. 当策略是确定性, 且动作空间是连续的, 我们便可进一步得到确定性策略梯度 (deterministic policy gradient, DPG) 定理 [76]:

$$\nabla_\theta V^{\pi_\theta}(s)=\mathbb{E}_{s\sim\mu^{\pi_\theta}(\cdot)}\left[\nabla_\theta\pi_\theta(a|s)\cdot\nabla_a Q^{\pi_\theta}(s,a)|_{a=\pi_\theta(s)}\right]. \tag{8}$$

关于策略梯度最经典的应用是 REINFORCE 算法 [77], 其使用累积奖赏 $R_t=\sum_{t'=t}^{T}\gamma^{t'-t}r_{t'}$ 作为对 $Q^{\pi_\theta}(s_t,a_t)$ 的估计来对策略参数进行更新.

2.1.4　基于行动者 – 评论者的强化学习方法

传统的行动者 – 评论者 (Actor-Critic) 方法由两个部分组成: Actor, 用以调整策略 π_θ 的参数 θ; 以及 Critic, 用来调整状态 – 动作值函数 $Q_w^{\pi_\theta}$ 的参数 w. 基于前述两类方法, 相应地也可以得到关于 Actor 和 Critic 的更新方式分别为

$$\begin{aligned}\theta\leftarrow&\theta+\alpha_\theta Q_w(s,a)\nabla_\theta\log\pi_\theta(s,a),\\ w\leftarrow&w+\alpha_w(r+\gamma Q_w(s',a')-Q_w(s,a))\nabla_w Q_w(s,a).\end{aligned} \tag{9}$$

Actor-Critic 方法将策略梯度方法和值函数近似的方法两者结合, Actor 基于概率选择动作, Critic 基于 Actor 选择的动作以及当前状态评判该动作的得分, 然后 Actor 根据 Critic 的评分修改其选择动作的概率. 其优势在于此类方法可以进行单步更新, 在连续动作空间中得到低方差的解 [78, 79], 而代价

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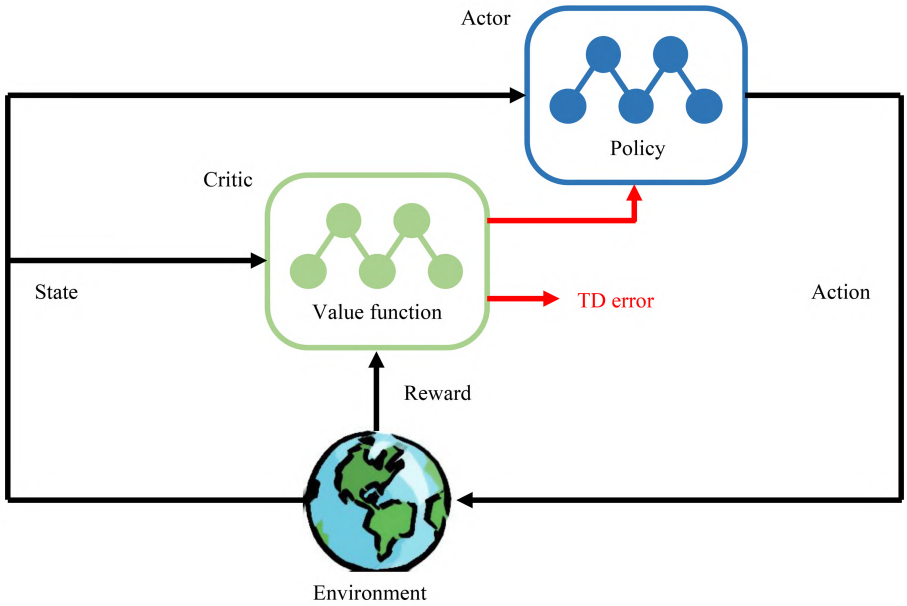

图 3　(网络版彩图) 行动者 – 评论者方法体系结构.

Figure 3　(Color online) Structure of actor-critic methods.

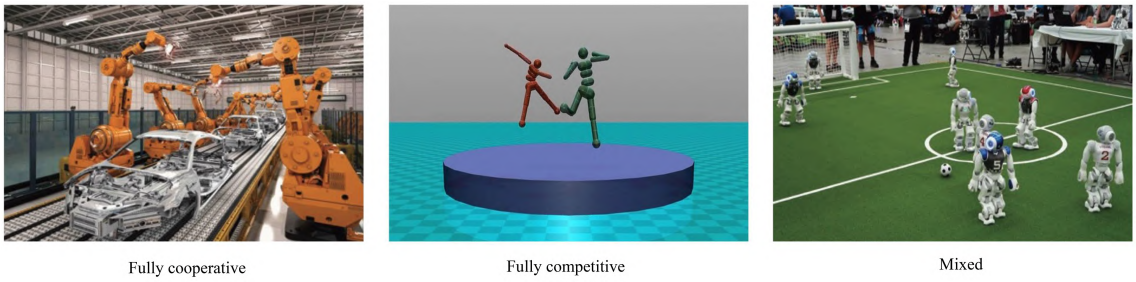

图 4　(网络版彩图) 多智能体系统的 3 种常见设定.

Figure 4　(Color online) Three common settings in multi-agent systems.

则是学习开始时, 由于 Critic 的估计不够准确, 算法有较大的波动. 图 3 描述了 Actor-Critic 方法的体系结构, 更多关于单智能体强化学习的方法可以参见综述 [9].

2.2　多智能体强化学习

2.2.1　多智能体系统

多智能体系统由分布式人工智能 (distributed artificial intelligence, DAI) 演变而来 [80], 其研究目的是解决大规模、复杂、实时和有不确定性的现实问题, 此类问题通过单智能体建模往往会效率低下并且与现实条件相悖. 多智能体系统具有自主性、分布性、协调性等特点, 并具有自组织能力、推理能力和学习能力. 自 20 世纪 70 年代被提出以来, 经过多年的发展, 目前多智能体系统的研究既包括构建单个智能体的技术, 如建模、推理、学习及规划等, 也包括使多个智能体协调运行的技术, 例如交互通信、协调、合作、协商、调度、冲突消解等. 多智能体系统在智能机器人、交通控制、分布式决策、软件开发、游戏等领域都得到迅速而广泛的应用, 目前已经成为一种对复杂系统进行分析与模拟的工具 [15].

如图 4 所示, 依据任务特性的不同, 一般可以将多智能体系统划分为完全协作 (fully cooperative setting)、完全竞争 (fully competitive setting) 和混合关系 (mixed setting) 这 3 种设定. 其中完全协作

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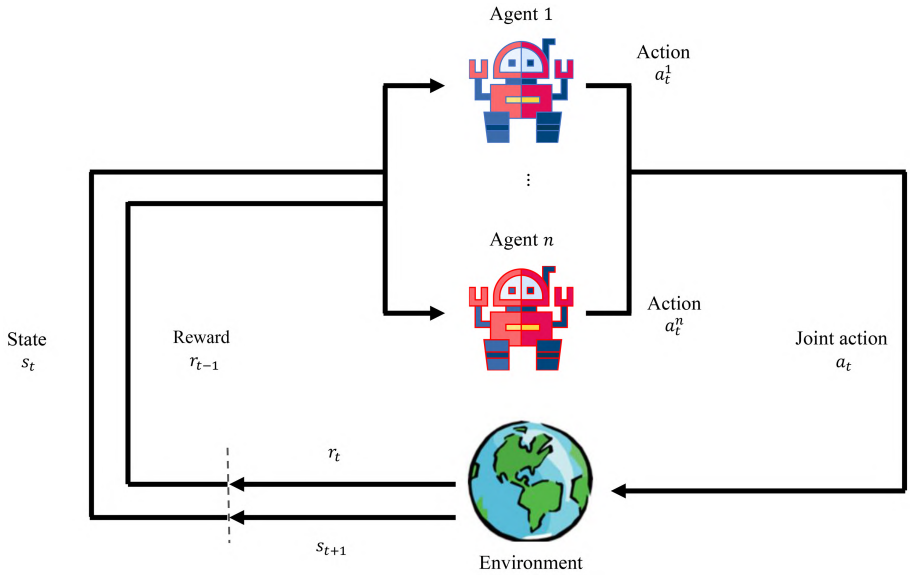

图 5　(网络版彩图) 多智能体强化学习示意图.

Figure 5　(Color online) Diagram of multi-agent reinforcement learning.

关系指系统中的智能体拥有一个相同的目标, 智能体相互协作完成特定任务, 例如机器系统装配; 完全竞争关系下, 智能体间的目标冲突, 一方的收益导致另一方的损失, 例如相扑对抗中的双方; 混合关系指智能体间可以划分为多个群组, 组内智能体相互合作, 而组间是相互竞争的关系, 例如足球比赛中, 队友间是协作关系而队与队之间是竞争关系.

多智能体系统中, 不同设定下的策略优化目标有差别, 其所学出的策略也会有所不同. 在协作的设定下, 每个智能体的决策要考虑到队友的策略, 与队友做到尽量好的配合, 避免产生干扰. 而在竞争的设定下, 智能体要考虑到对手的策略, 最小化对手同时最大化自己的回报, 并以此为目标相应优化自身策略.

2.2.2　多智能体强化学习问题设定

多智能体学习 (multi-agent learning, MAL) 将机器学习技术引入多智能体系统领域, 研究如何设计算法来创建动态环境下的自适应智能体学习与优化. 多智能体强化学习 (MARL) (如图 5 所示) 的目标是在共享环境中通过强化学习方法训练多个智能体, 从而完成给定的任务 [19, 56].

与单智能体强化学习中将多步决策过程建模为 MDP 不同, 多智能体强化学习一般被建模为随机博弈 (stochastic game) 模型.

定义2 (随机博弈)　随机博弈可以由一个六元组 $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}^i\}_{i\in\mathcal{N}}, P, \{R^i\}_{i\in\mathcal{N}}, \gamma\rangle$ 构成, 其中,

• $\mathcal{N} = \{1, 2, \ldots, n\}$ 代表系统中的智能体集合, 当 $n = 1$ 时, 问题退化成单智能体马尔可夫决策过程, $n \geqslant 2$ 是一般的随机博弈过程;

• $\mathcal{S}$ 是环境中所有智能体共享的状态空间;

• $\mathcal{A}^i$ 是智能体 i 的动作空间, 定义联合动作空间 $\mathcal{A} := \mathcal{A}^1 \times \cdots \times \mathcal{A}^n$;

• $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ 是状态转移函数, 具体而言, 它刻画了在每个时间步, 给定一个联合动作 $\boldsymbol{a} \in \mathcal{A}$, 环境从 $s \in \mathcal{S}$ 转移到另外一个状态 $s' \in \mathcal{S}$ 的概率;

• $R^i : \mathcal{S} \times \mathcal{A} \times \mathcal{N} \to \mathbb{R}$ 是每个智能体的奖赏函数;

• $\gamma \in [0, 1]$ 是折扣因子.

在每一时间步, 智能体 $i \in \mathcal{N}$ 处于状态 s, 选择动作 $a^i \in \mathcal{A}^i$, 构成联合动作 $\boldsymbol{a} = \langle a^1, \ldots, a^n\rangle$ 并执行, 环境转移到下一个状态 $s' \sim P(\cdot \mid s, \boldsymbol{a})$, 智能体 i 获得自身奖赏 $R^i(s, \boldsymbol{a})$. 每个智能体通过优化自

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身的策略函数 $\pi_i : \mathcal{S} \to \Delta(\mathcal{A}^i)$ 以最大化其期望累积奖赏, 以状态价值函数的形式表达如下:

$$\max_{\pi_i} V^i_{\boldsymbol{\pi}}(s) := \mathbb{E}\left[\sum_{t\geqslant 0} \gamma^t R^i(s_t, \boldsymbol{a}_t) | a^i_t \sim \pi_i(\cdot \mid s_t), \boldsymbol{a}^{-i}_t \sim \boldsymbol{\pi}_{-i}(\cdot \mid s_t), s_0 = s\right], \tag{10}$$

其中符号 $-i$ 代表除了智能体 i 之外的其他智能体. 与单智能体强化学习中智能体仅仅需要考虑自身对环境的影响不同, 多智能体系统中智能体之间也会相互影响, 他们共同决策并且同时更新策略. 当系统中的其他智能体的策略固定时, 智能体 i 可以最大化自己的收益函数, 以寻找到相对于其他智能体的策略的最优策略 π^*_i. 在多智能体强化学习中, 合理性 (rationality) 与收敛性 (convergence) 是学习算法最主要的评价指标. 具体地, 我们称一个算法具备合理性, 如果智能体在对手使用固定策略的情况下, 能够基于该算法学习到相应的最优策略. 当系统中其他智能体也同样基于算法进行策略学习时, 如果能够收敛到稳定的策略, 则进一步可称该算法具有收敛性.

由上述讨论内容可知, 在多智能体强化学习的过程中, 每个智能体希望增加自身的效用, 此时的学习目标可以是最大化自身的 Q- 函数. 因此一个直观的学习方法是对每个智能体构建一个独立的 Q- 函数, 并按照 Q- 学习算法进行更新:

$$Q_i(s_t, a^i_t) \leftarrow Q_i(s_t, a^i_t) + \alpha\left[r_{t+1} + \gamma \max_{a^i_{t+1}\in\mathcal{A}^i} Q_i(s_{t+1}, a^i_{t+1}) - Q_i(s_t, a^i_t)\right]. \tag{11}$$

与单智能体强化学习算法一样, 每个智能体通过贪心策略选择当前执行动作:

$$\pi_i(a^i_t | s_t) = \mathbb{1}\left\{a^i_t = \arg\max_{a^i_t} Q_i(s_t, a^i_t)\right\}. \tag{12}$$

上述情况中我们默认每个智能体都能够获取全局状态信息, 从而进行决策. 但是受限于现实条件, 多智能体中的个体往往只能获得有限的局部观测. 针对这一特性, 一般将这样的决策过程建模为部分可观测随机博弈 (partially-observable stochastic games, POSG).

定义3 (部分可观测随机博弈)　定义为 $\mathcal{M} = \langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}^i\}_{i\in\mathcal{N}}, P, \{R^i\}_{i\in\mathcal{N}}, \gamma, \{\Omega^i\}_{i\in\mathcal{N}}, \mathcal{O}\rangle$, 其前 6 项和随机博弈的定义 2 一致, 除此之外, 增加以下部分:

- Ω^i 是智能体 i 的观测空间, 所有智能体的联合观测空间为 $\boldsymbol{\Omega} := \Omega^1 \times \cdots \times \Omega^n$,
- $\mathcal{O} : \mathcal{S} \to \Delta(\boldsymbol{\Omega})$ 代表了观测函数, $\mathcal{O}(\boldsymbol{o}|s)$, 指在给定状态 s 后关于联合观测 $\boldsymbol{o}$ 的概率函数.

在部分可观测随机博弈中, 每个智能体通过更新自己的策略 $\pi_i \in \Pi : \Omega^i \to \Delta(\mathcal{A}^i)$ 以最大化收益. 虽然部分可观测随机博弈在真实场景中应用比较广泛, 然而理论结果证明其求解难度是 NEXP-hard 的 [81]. 所幸的是, 近几年提出的包括集中式的多智能体强化学习训练框架, 利用循环神经网络编码历史信息 [82] 和智能体通信 [20] 等技术可以从不同角度缓解该问题.

特别地, 在协作型任务中, 可以进一步将部分可观测随机博弈建模为分布式局部可观测马尔可夫决策过程 (decentralized partially observable MDP, Dec-POMDP) [81], 两者的主要区别在于后者中每个智能体的奖赏函数相同. 除了部分可观测这一困难外, 相较于单智能体强化学习, 多智能体强化学习还面临着更多独有的问题. 多智能体强化学习中, 状态空间变大, 动作空间随着智能体数量的增长出现 "维度爆炸" (curse of dimensionality) 问题 [56]. 此外, 可扩展性、信度分配、异质性、协同探索等问题的存在长期以来也在阻碍着多智能体强化学习的发展 [55].

2.2.3　多智能体强化学习训练范式

在多智能体强化学习中, 训练指通过获取的经验 (状态、动作、奖赏等) 对策略进行优化的过程, 执行指的是智能体根据自身或联合策略执行动作与环境交互. 一般而言, 根据智能体更新自身策略时是否需要其他智能体的信息, 多智能体强化学习的训练过程可以划分为集中式训练 (centralized training)

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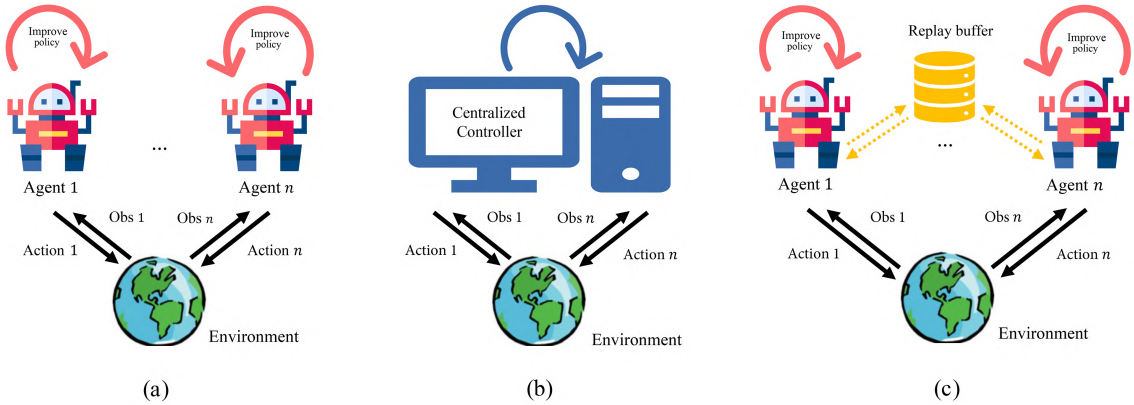

图 6　(网络版彩图) 三种不同的多智能体强化学习训练范式. (a) 分布式训练分布式执行; (b) 集中式训练集中式执行; (c) 集中式训练分布式执行.

Figure 6　(Color online) Three different MARL training paradigms. (a) Decentralized training decentralized execution; (b) centralized training centralized execution; (c) centralized training decentralized execution.

与分布式训练 (decentralized training); 与此相对应的是, 根据在执行阶段是否需要外部信息, 分为集中式执行 (centralized execution) 与分布式执行 (decentralized execution). 结合训练阶段与执行阶段, 多智能体强化学习可以分为 3 种范式: 分布式训练分布式执行 (decentralized training decentralized execution, DTDE)、集中式训练集中式执行 (centralized training centralized execution, CTCE)、集中式训练分布式执行 (centralized training decentralized execution, CTDE). 各结构如图 6 所示. 在分布式训练分布式执行 (DTDE) 框架下, 每个智能体仅利用自己的局部信息独立地进行策略更新和策略实现, 不涉及信息的交换, 智能体策略可表示为 $\pi_i : \Omega^i \to \Delta(\mathcal{A}^i)$. IQL [83] 是基于 DTDE 的一个典型算法, 由于此类算法仅需要考虑某个智能体本身, 具有较强的可扩展性; 同时, 由于智能体不考虑其他智能体的信息或状态, 智能体往往处于非稳态过程, 无法学到最优的策略. 为了解决分布式学习带来的效率低下问题, 乐观优化与滞后优化通过对目前较好的 Q 赋予更大的更新权重, 在一定程度上能缓解以上问题. 与此同时, 在通过 DQN 进行策略更新时, 由于其他智能体的策略也进行了更新, 直接使用经验回放池的数据会加剧非稳态, 对此, 可以通过重要性采样缓解此问题:

$$\mathcal{L}(\theta_i) = \frac{\boldsymbol{\pi}_{-i}^{t_c}(\boldsymbol{a}^{-i}|\boldsymbol{o}^{-i})}{\boldsymbol{\pi}_{-i}^{t_i}(\boldsymbol{a}^{-i}|\boldsymbol{o}^{-i})}[(y_i^Q - Q_i(o^i, a^i; \theta_i))^2],$$

其中 θ_i 为智能体 i 的 Q- 网络参数, t_c 为当前时刻, t_i 是样本的采集时间, y_i^Q 是时序差分目标, 计算与单智能体部分所介绍类似. 与此同时, 使用循环神经网络等技术也可以在一定程度上缓解上述问题. 相较于分布式训练, 集中式训练框架允许智能体在训练过程中进行信息交换, 甚至可以拿到其他智能体的所有信息.

相对于 DTDE, 集中式训练集中式执行则允许所有智能体可以观测到全局信息, 以 $\boldsymbol{\pi} : \boldsymbol{\Omega} \to \Delta(\boldsymbol{\mathcal{A}})$ 的形式同时进行决策.

在 CTCE 框架下, 可以使用任意一种单智能体强化学习算法训练多智能体系统. 在该类框架中, 算法的复杂度随着状态和动作的维度增长呈现维度爆炸 [84], 该问题一般可以通过策略分解 (例如 $\boldsymbol{\pi}(\boldsymbol{a}|\boldsymbol{o}) = \prod_{i=1}^n \pi_i(a^i|o^i)$) 或者值分解 (例如 $Q_{\text{tot}}(o^1, \ldots, o^n, a^1, \ldots, a^n) = \sum_i Q_i(o^i, a^i)$) 来解决. 虽然分解类方法可以缓解维度爆炸问题, 但是 CTCE 无法评估每个智能体间的相互影响, 会给学习效率带来较严重的影响. 近年来随着多智能体强化学习协作理论的发展, 集中式训练分布式执行框架得到广泛关注并取得重大成功.

在集中式训练分布式执行 (CTDE) 框架下, 在训练阶段, 智能体通过结合其他智能体的信息甚至是全局信息以优化自己的局部策略: $\pi_i : \Omega^i \to \Delta(\mathcal{A}^i)$. 而在分布式执行过程中, 智能体仅使用自己的局

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部信息进行决策. CTDE 框架近年来在多智能体系统中得到广泛应用, 尤其在某些复杂的场景下, 更是比前两种范式有更明显的优势. 在训练过程中, 拿到全局信息, 可以缓解非稳态性; 在执行过程中, 智能体可以直接基于局部信息按策略执行动作. 尽管 CTDE 取得了较好的结果, 但是目前实验发现, CTDE 在处理异质多智能体 (智能体的状态或者动作空间不一致) 时, 往往表现不佳, 为了解决这些问题, 可以通过技能学习或者通过先分组, 再采用局部 CTDE 的方式进行训练, 进一步, 一些研究工作也从其他方面对集中式与分布式进行讨论 [85, 86].

2.2.4　多智能体强化学习的难点与挑战

相较于单智能体强化学习, 真实社会中的场景往往更适合建模为多智能体强化学习, 然而由于多个智能体的存在, 各个智能体的策略同时在更新, 往往会带来更多的挑战. 本部分对多智能体强化学习中存在的主要挑战包括非稳态 (non-stationarity)、可扩展性 (scalability)、部分可观测 (partial observability) 及目前存在的解决方法进行介绍.

在单智能体系统中, 智能体只需考虑本身与环境的交互即可, 此时, 环境的转移是固定的. 然而在多智能体系统中, 环境中多个智能体同时进行策略更新, 这样带来的结果是, 站在特定智能体的视角, 将其他智能体视作环境的一部分, 这样的环境就是非稳态 [21] 的, 是一个动态目标过程. 即当智能体 i 的策略由 π_i 改变为 $\bar{\pi}_i$ 时, 对于除此之外的智能体 $\boldsymbol{\pi}_{-i}$ 而言, 环境的状态转移函数也发生改变 $P(s'|s, \boldsymbol{a}, \pi_i, \boldsymbol{\pi}_{-i}) \neq P(s'|s, \boldsymbol{a}, \bar{\pi}_i, \boldsymbol{\pi}_{-i})$.

非稳态表明, 随着环境中多个智能体的同时学习与更新, 学习不再遵守马尔可夫性. 非稳态问题在基于 Q- 更新的问题中更严重, 尤其是依靠经验回放池更新的一类方法, 由于进行 TD 更新需要进行动作采样, 在多智能体中, 采样未来的联合动作往往难度较大; 另一方面, 随着所有智能体同时进行更新, 经验回放池中的数据可能会过时. 解决环境非稳态可以通过对手 (队友) 建模, 对回放数据进行重要性采样, 进行集中式训练或者采用元强化学习考虑到其他队友的更新等方法. 而目前的多智能体方法为解决非稳态问题, 一般会考虑环境中其他智能体的动作, 带来可扩展性问题, 即随着智能体的数量增多, 联合动作空间大小呈指数上升. 真实场景中的多智能体往往数量较多, 比如自动驾驶场景中, 存在成百上千甚至更多的智能体, 可扩展性问题的解决是有必要的但也是极具挑战性的. 目前的方法包括参数共享 (同质智能体共享神经网络, 异质智能体间独立训练) [24]; 或者通过迁移学习或者课程学习, 从少到多进行训练, 最终迁移到智能体较多的场景 [60].

考虑到环境中传感器的限制等因素, 智能体往往很难获得全局状态, 一般只能拿到部分信息, 我们一般把多智能体强化学习建模为 POSG (见定义 3), 因为智能体只能拿到局部观测信息, 从单个智能体的角度, 环境不再遵守马尔可夫性, 会给训练带来困难. 针对上述问题, 目前的解决方案包括多智能体通信: 智能体之间通过信息传递, 缓解多智能体的局部观测问题. 通信需要解决的以下问题: 与谁通信, 通信什么内容, 何时通信. 对第一个问题, 目前包括三组拓扑结构如图 7 所示, 每个智能体将信息广播 (broadcast) 给环境中所有的智能体; 只发送给特定智能体, 把智能体交互建模为网络结构; 只与部分邻居进行通信. 从通信内容上来看, 直接把自身局部信息发送给其他智能体是最直接的方式, 然而此种通信方式往往会带来带宽损耗, 也存在信息冗余; 另外一种方式是发送者对局部信息先进行处理, 提取出最有用的信息; 针对通信时机, 可以在每时每刻都通信, 或者在关键时刻进行通信, 目前的研究发现, 一直通信会带来不必要的信息冗余, 如果只在关键时刻通信, 通过学到稀疏的通信结构, 效率往往会更高.

3　经典环境下的协作多智能体强化学习

3.1　多智能体协作概念

第 2.2.1 小节提到多智能体系统根据任务特性可以划分为协作型、竞争型与混合型这 3 种设定,

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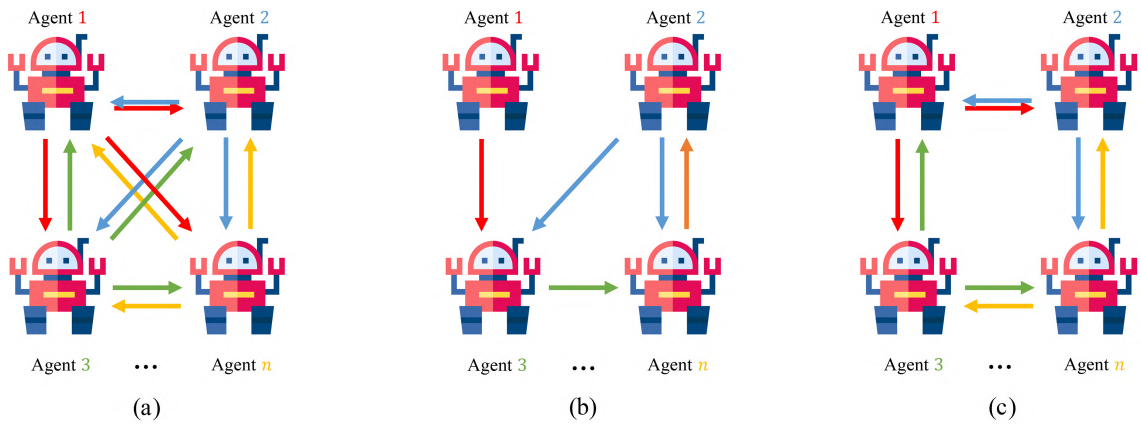

图 7　(网络版彩图) 三种不同的多智能体通信拓扑. (a) 广播通信; (b) 针对性通信; (c) 仅与邻居通信.

Figure 7　(Color online) Three different multi-agent communication topologies. (a) Broadcast communication; (b) targeted communication; (c) neighbor-only communication.

其中竞争型一般通过建模为零和博弈进行求解, 混合型是合作型与竞争型的混合, 本部分重点介绍了目前研究较深入的协作型多智能体强化学习算法.

定义4 (完全协作)　多智能体强化学习中的完全协作指所有智能体共享一个奖赏函数, 满足

$$R^1 = R^2 = \cdots = R^n = R.$$

在协作型多智能体强化学习中, 因为智能体共享一个相同的奖赏函数, 我们很容易知道智能体间拥有一个相同的回报. 由于智能体共享相同的奖赏, 如果存在一个中心化的控制器, 那么在训练过程中, 多智能体强化学习会退化成一个马尔可夫过程, 智能体的动作空间会变成一个随机博弈的联合动作空间, 其优化目标如下:

$$Q(s_t, \boldsymbol{a}_t) \leftarrow Q(s_t, \boldsymbol{a}_t) + \alpha \left[r_t + \gamma \max_{\boldsymbol{a}_{t+1} \in \boldsymbol{\mathcal{A}}} Q(s_{t+1}, \boldsymbol{a}_{t+1}) - Q(s_t, \boldsymbol{a}_t) \right]. \tag{13}$$

在真实世界中, 上式难以直接作为多智能体决策的优化目标, 这是因为在多智能体系统中智能体往往是独立进行决策的. 具体而言, 在特定的状态下, 可能会存在多组最优联合动作, 因为分布式执行的存在, 各个智能体可能会偏向于选择不同的最优动作, 也就会导致协作行为的涌现. 一种最直接的做法是忽略智能体间的协作行为 [87], 我们假设仅存在一个最优均衡点, 这样可以直接使用上述公式. 早期的分布式 Q- 学习算法在全局可观测的假设下, 直接忽略智能体间的协作, 智能体只更新自己的局部动作价值函数 $Q_i(s, a^i)$.

定义5 (免协作类多智能体协作方法)　假设每个智能体的局部动作价值函数为 $Q_i(s_t, a_t^i)$, 则其更新为

$$Q_i(s_t, a_t^i) \leftarrow \max \left\{ Q_i(s_t, a_t^i),\ r_t + \gamma \max_{a_{t+1}^i \in \mathcal{A}^i} Q_i(s_{t+1}, a_{t+1}^i) \right\}. \tag{14}$$

免协作类多智能体协作方法可以忽略掉奖赏较差的动作, 然而一直取最大操作可能会带来过高估的问题, 另外一方面, 因为动作价值的计算基于联合动作函数, 往往扩展性较差. 与之相对的是, 我们可以考虑协作方式, 采用联合学习方式进行策略优化 [88].

定义6 (联合动作学习)　假设智能体处于全局可观测的环境中, 可以观测到所有队友的状态和动作, 每个智能体的策略为 π_i, 存在一个全局价值函数 $Q(s, \boldsymbol{a})$, 那么对智能体 i, 可以评估其局部动作价值为

$$Q_i(s_t, a_i) =: \sum_{\boldsymbol{a}_{-i} \in \boldsymbol{\mathcal{A}}_{-i}} Q(\langle a_i, \boldsymbol{a}_{-i} \rangle) \prod_{j \neq i} \pi_j(a_j).$$

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与此同时, 在特殊场景下, 可以通过角色分配、协作图、多智能体通信等方式间接实现多智能体协作. 与此同时, 分布式执行系统往往只能得到局部观测, 一般将协作型多智能体强化学习建模为分布式部分可观测马尔可夫决策过程 DEC-POMDP. DEC-POMDP 是 POSG 的一种特殊情况, 其中各个智能体拥有相同的奖赏函数. 我们发现, 如果所有智能体共享奖赏函数和回报, 某些智能体不需要对系统作出实质贡献也可以获得奖赏. 因此, 在协作多智能体系统中, 解决准确评估各个智能体的行为对整个系统的贡献这一信度分配问题 [22,89] 也具备一定的挑战性.

3.2　典型协作方法

3.2.1　值分解的协作方法

与单智能体强化学习类似, 基于值分解的多智能体强化学习方法利用 Q- 函数指导动作的选择, 在星际争霸微操任务 [33] 等复杂环境中有良好的表现. 为缓解部分可观测性带来的问题, 值函数的输入一般为轨迹 $\tau_t^i = (o_1, a_1, \ldots, a_{t-1}, o_t) \in \mathcal{T}^i$.

值函数分解的方法大多建立在个体 – 全局 – 最大原则 (individual-global-max, IGM) 的基础上 [90].

定理1 (个体 – 全局 – 最大原则)　对于一个联合动作价值函数 $Q_{\text{tot}} : \boldsymbol{\mathcal{T}} \times \boldsymbol{\mathcal{A}}$ 以及局部个体动作价值函数 $\{Q_i : \mathcal{T}^i \times \mathcal{A}^i \to R\}_{i=1}^n$, 下式成立:

$$\arg\max_{\boldsymbol{a}} Q_{\text{tot}}(\boldsymbol{\tau}, \boldsymbol{a}) = \begin{pmatrix} \arg\max\limits_{a^1} Q_1(\tau^1, a^1) \\ \vdots \\ \arg\max\limits_{a^n} Q_n(\tau^n, a^n) \end{pmatrix}.$$

则称 Q_{tot} 与 $\{Q_i\}_{i=1}^n$ 满足个体 – 全局 – 最大原则.

目前满足 IGM 的典型值分解方法有 3 种. VDN [30] 用局部 Q- 函数的加和表示联合 Q- 函数:

$$Q_{\text{tot}}^{\text{VDN}}(\boldsymbol{\tau}, \boldsymbol{a}) = \sum_{i=1}^{n} Q_i\left(\tau^i, a^i\right). \tag{15}$$

这样的值分解方法虽然满足 IGM, 但是其网络表达能力不足, 在复杂场景下往往表现不佳, QMIX [31] (见图 8) 通过扩展 VDN, 引入混合网络, 结合每个智能体的 Q- 值与全局状态计算全局效用, 并进行信度分配. 为保证 IGM, 混合网络 (mixing network) 将个体 Q- 值与状态作为输入, 得到全局 Q- 值并满足一定条件:

$$\begin{aligned} Q_{\text{tot}}^{\text{QMIX}}(\boldsymbol{\tau}, \boldsymbol{a}) =& \text{Mixing}(s, Q_1(\tau^1, a^1), \ldots, Q_n(\tau^n, a^n)), \\ & \forall i \in \mathcal{N}, \frac{\partial Q_{\text{tot}}^{\text{QMIX}}(\boldsymbol{\tau}, \boldsymbol{a})}{\partial Q_i\left(\tau^i, a^i\right)} \geqslant 0. \end{aligned} \tag{16}$$

在优化时, 计算时序差分损失以进行优化:

$$\mathcal{L}_{\text{QMIX}} = (Q_{\text{tot}}^{\text{QMIX}}(\boldsymbol{\tau}, \boldsymbol{a}) - y_{\text{tot}})^2, \tag{17}$$

其中 $y_{\text{tot}} = r + \gamma \max_{\boldsymbol{a}'} Q_{\text{tot}}^{\text{QMIX},-}(\boldsymbol{\tau}', \boldsymbol{a}')$, 以及 $Q_{\text{tot}}^{\text{QMIX},-}$ 是目标 Q- 网络. 基于 QMIX 的值分解方法凭借其表达能力与网络结构之间的相对简略, 在各种环境上取得较优的结果, 其伪代码见算法 1.

VDN 和 QMIX 设计所满足的条件是 IGM 性质的充分不必要条件, 导致其在表达能力上仍有一定的欠缺. 为了进一步增强神经网络的表达能力, QPLEX [91] 提出一种新的双重对决多智能体 (duplex dueling multi-agent, DDMA) 结构:

$$Q_{\text{tot}}^{\text{QPLEX}}(\boldsymbol{\tau}, \boldsymbol{a}) = V_{\text{tot}}(\boldsymbol{\tau}) + A_{\text{tot}}(\boldsymbol{\tau}, \boldsymbol{a}) = \sum_{i=1}^{n} Q_i\left(\boldsymbol{\tau}, a^i\right) + \sum_{i=1}^{n} \left(\lambda^i(\boldsymbol{\tau}, \boldsymbol{a}) - 1\right) A_i\left(\boldsymbol{\tau}, a^i\right), \tag{18}$$

其中 $\lambda^i(\boldsymbol{\tau}, \boldsymbol{a})$ 是通过多头注意力机制 [92] 计算, 得到对不同的优势函数进行的信度分配. QPLEX 凭借其表达能力与网络结构, 在多种环境下都取得了较优的结果.

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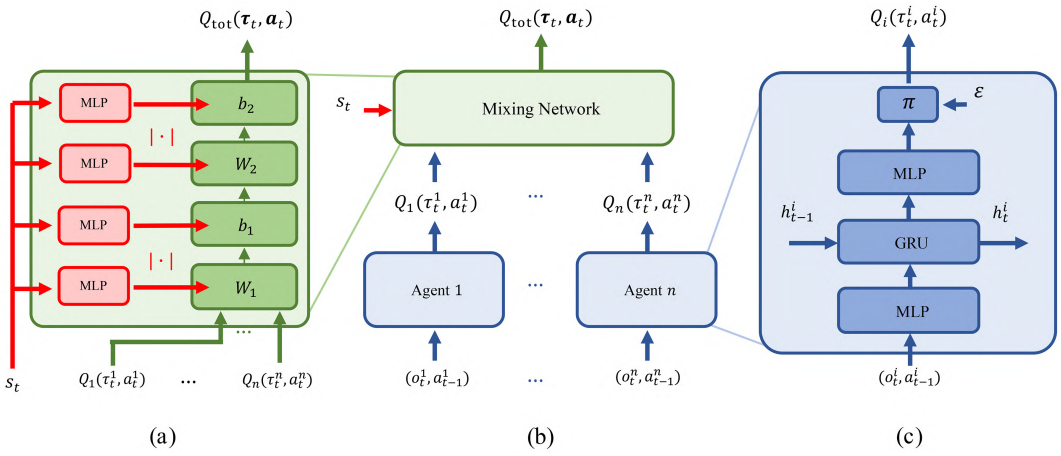

图 8　(网络版彩图) QMIX 算法框架图. (a) 混合网络; (b) 整体架构; (c) 智能体 Q- 网络.

Figure 8　(Color online) Framework of QMIX algorithm. (a) Mixing network; (b) overall architecture; (c) agent Q-network.

算法 1 QMIX 算法.

初始化网络参数 θ, 包括混合网络、智能体的局部神经网络、超网络;
设置学习率 α, 初始化经验回放池 $D = \emptyset$;
while 训练未结束 **do**
　令 $t = 0$, s_0 为初始状态;
　初始化轨迹 $\{\tau_{-1}^i = \emptyset\}_{i \in \mathcal{N}}$;
　while s_t 未到终止状态并且 t 小于最大轨迹长度 **do**
　　for 每个智能体 i **do**
　　$\tau_t^i = \tau_{t-1}^i \cup \{o_t^i\}$;
　　$\epsilon = \text{epsilon_schedule}(t)$;
　　$a_t^i = \begin{cases} \arg\max_{a_t^i} Q(\tau_t^i, a_t^i), & \text{with probability } 1-\epsilon, \\ \text{Randint}(1, |\mathcal{A}^i|), & \text{with probability } \epsilon; \end{cases}$
　　$\tau_t^i = \tau_t^i \cup \{a_t^i\}$;
　　end for
　　执行联合动作后, 获得奖赏函数 r_t 和下一时刻的状态 s_t;
　　令 $D = D \cup \{(s_t, \boldsymbol{a}_t, r_t, s_{t+1})\}$;
　　$t = t + 1$;
　end while
　if $|D| \geqslant \text{batch_size}$ **then**
　　$b \leftarrow$ 随机从 D 中采样;
　　for b 中的每条轨迹的每个转移样本 **do**
　　通过式 (16), 计算 $Q_{\text{tot}}^{\text{QMIX}}$ 与 $y_{\text{tot}}^{\text{QMIX}}$;
　　end for
　　通过式 (17) 计算 $\mathcal{L}_{\text{QMIX}}$ 进行优化;
　end if
　if 更新时间超过设计的阈值 **then**
　　更新目标网络;
　end if
end while

3.2.2　基于策略梯度的多智能体协作

基于策略梯度的多智能体强化学习方法用价值函数评估策略的性能, 各个智能体间的奖赏与回报相同, 使得所有智能体拥有相同的价值函数, 记全局的动作价值函数为 $Q_{\boldsymbol{\pi}}(s, \boldsymbol{a})$, 状态价值函数为 $V_{\boldsymbol{\pi}}(s)$, 这里值得注意的是, 在多智能体强化学习中某个智能体的动作价值函数与状态价值函数依赖于

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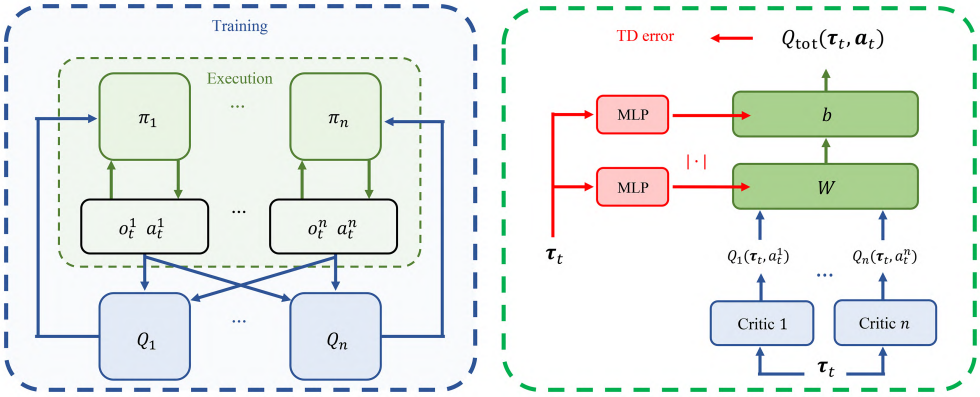

(a)　　(b)

图 9　(网络版彩图) (a) 多智能体深度确定性策略梯度与 (b) 可分解的异策略多智能体策略梯度算法结构图.

Figure 9　(Color online) Algorithm frameworks of (a) multi-agent deep deterministic policy gradient and (b) off-policy multi-agent decomposed policy gradient.

所有多智能体的策略:

$$\pi_1(a^1|o^1;\theta^1), \pi_2(a^2|o^2;\theta^2), \ldots, \pi_n(a^n|o^n;\theta^n),$$

其中 θ^i 是智能体 i 的策略参数, 智能体间的策略差异也主要体现在策略参数的差异上. 在策略学习过程中, 我们的目的是通过优化策略参数以最大化目标函数:

$$J(\theta^1, \ldots, \theta^n) = E_s[V_{\boldsymbol{\pi}}(s)].$$

所有智能体目标一致, 都是通过优化自己的策略参数 θ^i 以最大化目标函数 J, 因此, 在该目标下对参数做梯度上升可以实现策略优化.

定理2 (协作型 MARL 的策略梯度定理)　假设存在一个与联合动作无关的基线函数 b, 则协作型 MARL 存在如下的策略梯度:

$$\nabla_{\theta^i} J(\theta^1, \ldots, \theta^n) = \mathbb{E}[\nabla_{\theta^i} \log \pi_i(a^i|o^i;\theta^i) \cdot (Q_{\boldsymbol{\pi}}(s, \boldsymbol{a}) - b)], \tag{19}$$

其中联合动作从联合动作概率 $\boldsymbol{\pi}(\boldsymbol{a}|\boldsymbol{o}, \boldsymbol{\theta}) = \pi_1(a^1|o^1;\theta^1) \times \cdots \times \pi_n(a^n|o^n;\theta^n)$ 中采样.

这里重点介绍多智能体深度确定性策略梯度 (multi-agent deep deterministic policy gradient, MADDPG) [28] (图 9), 每个智能体对应一个分布式的确定性策略 Actor 模块 $\pi_i(o^i;\theta^i)$, 智能体根据 Actor 进行决策; 另一方面, 每个智能体配备一个集中式的 Critic 网络 $Q_i(s, \boldsymbol{a};\phi^i)$ 用以指导更新优化 Actor 网络. 具体而言, MADDPG 会另外维护延迟更新的目标策略 $\pi_i'(o^i;\theta_-^i)$ 用以计算 Critic 的时序差分目标值. 从而得到 Critic 的优化目标:

$$\begin{aligned} y_i^Q &= r + \gamma Q_i(s', \boldsymbol{a}'|\phi^i)|_{a'^j = \pi_j'(o'^j;\theta_-^j)}, \\ \mathcal{L}(\phi^i) &= \frac{1}{2}(y_i^Q - Q_i(s, \boldsymbol{a};\phi^i))^2. \end{aligned} \tag{20}$$

进一步, 用 Critic 指导 Actor 进行如下更新:

$$\nabla_{\theta^i} J(\theta^i) = \nabla_{\theta^i} \log \pi_i(o^i;\theta^i) \nabla_{a^i} Q_i(s, \boldsymbol{a};\phi^i)|_{a^j = \pi_j(o^j|\theta^j)}. \tag{21}$$

MADDPG 是一个典型的基于 CTDE 框架的算法, Critic 网络的训练需要拿到环境的全局状态信息, 在拿不到全局状态信息的环境中, 往往需要拿到其他所有智能体的观测信息 o_i, 当智能体数量增

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多, 往往会加大 Critic 网络的训练难度, 一般可以采用基于注意力机制 [93] 等方式进行信息聚合, 缓解由于数量变化带来的计算代价. 另外一方面, MADDPG 可以处理竞争的场景, 可能由于隐私等因素, 往往无法拿到其他智能体的动作, 可以通过对手建模 [59] 实现对其他智能体的动作信息进行估计. 我们可以发现, MADDPG 在协作型多智能体强化学习任务中, 没有显式考虑信度分配问题, 为了建模协作型多智能体系统中各个智能体的贡献 (或者应该分得的奖赏), 反事实基线多智能体策略梯度算法 (counterfactual multi-agent policy gradients, COMA) [89] 得到较为广泛的应用. COMA 基于反事实基线 $A_i(s,\boldsymbol{a}) = Q(s,\boldsymbol{a}) - \sum_{a'^i} \pi^i(a'^i|o^i)Q(s,\langle \boldsymbol{a}^{-i}, a'^i\rangle)$. 代替式 (19) 中的 Q- 函数减去基线函数项, 即可得到更加学习高效的策略梯度. 需要注意的是, 在基于策略梯度的方法中, 同样可以使用轨迹代替局部观测作为输入提高学习效率与稳定性.

3.2.3　结合策略梯度与值分解的多智能体协作

除了前述基于策略梯度与值分解类的方法之外, 也有部分工作通过集合两者的优势以开发算法, 其中可分解的异策略多智能体策略梯度 (off-policy multi-agent decomposed policy gradient, DOP) [94] (如图 9 所示), 将值分解的思想引入多智能体行动者 – 评论者框架, 并学习一个集中但分解的评论者. 该框架将集中式评论者分解为以局部动作为输入的个体评论者的加权线性求和形式. 这种分解结构不仅实现了评论者的可扩展学习, 还带来了几个好处. 它实现了对随机策略的可行异策略评估, 减轻了集中式分布式不一致问题, 并隐式地学习了高效的多智能体奖赏分配. 基于这种分解, DOP 为离散和连续动作空间开发了高效的异策略多智能体分解策略梯度方法.

DOP 的具体贡献包括: (1) 为了解决多智能体的奖赏分配以及学习集中式评论者的可扩展性问题, DOP 利用线性值函数分解学习集中化的评论者, 这样可以本地推导出每个智能体的基于个体值函数的策略梯度; (2) DOP 提出随机策略梯度和确定性策略梯度多智能体强化学习算法, 分别有效执行离散动作任务和连续动作任务; (3) 考虑策略梯度算法的样本效率问题, DOP 结合线性值函数分解和树回溯 (tree backup) 技术, 提出可扩展的异策略评论者学习方法, 从而提高策略学习的样本效率. 除此之外, 也有其他的工作从其他方面结合值分解与策略梯度之间的优点, 其中 VDAC [95] 直接将策略梯度函数类方法与值分解方法进行结合, 提出基于线性求和与基于混合神经网络的方法. IAC [96] 在 MAAC 的基础上通过值分解方法进一步优化智能体间的信度分配, 在多个环境上取得令人惊喜的协作效果. FACMAC [97] 提出可分解联合策略梯度的多智能体算法, 并对应提出针对连续环境下的多智能体测试环境 MAMuJoCo; FOP [98] 设计一种基于最大熵的值分解与梯度分解方法; 另一方面, RIIT [99] 将当前多智能体强化学习中的一些常见技巧整合进当前的协作算法中, 并针对性提出设计新的开源算法框架 PyMARL2 1).

3.3　经典环境下的协作多智能体强化学习分类

除了 3.2 小节涉及的协作能力提升类方法之外, 研究者们也从其他方面对协作多智能体进行研究, 主要包括高效探索、通信等方面, 各方面研究内容及代表工作如表 1 [28~34, 91, 94, 100~142] 所示.

3.3.1　多智能体协作探索

强化学习方法追求高效地学习到最优策略, 其中高质量的训练样本必不可少, 探索则在采样过程中扮演着关键角色, 是强化学习的一个重要环节 [143]. 如在单智能体场景中得到广泛应用的很多探索技术一样, 多智能体强化学习的探索也在多方面得到关注. 原始的多智能体强化学习算法, 如 QMIX 与 MADDPG 没有专门针对多智能体任务进行特殊探索设计, 在部分复杂场景下表现不佳, 由此针对多智能体探索的工作相继出现. MAVEN [101] 引入一个隐变量并最大化该隐变量与所产生轨迹之间的互信息以改善基于值分解类方法, 如 QMIX 等的探索能力不足的问题. EITI 与 EDTI [102] 通过考

1) https://github.com/hijkzzz/pymarl2.

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表 1　经典环境下的协作多智能体强化学习研究内容.

Table 1　Research content on cooperative multi-agent reinforcement learning in classical environments.

Research direction	Core content	Representative algorithms	Applications and achievements
Algorithm framework design	Utilize multi-agent coordination theories or design neural networks to enhance coordination	VDN [30], QMIX [31], QPLEX [91], MADDPG [28], MAPPO [29], HAPPO [100], DOP [94], MAT [32]	Showcasing good coordination performance in tasks such as SMAC [33] and GRF [32], demonstrating significant potential
Cooperative exploration	Design efficient exploration mechanisms to improve sampling and training efficiency	MAVEN [101], EITI (EDTI) [102], EMC [103], CMAE [104], UneVEn [105], SMMAE [106]	Significantly enhance cooperation effectiveness and address the issue of low collaboration capability in environments with sparse rewards
Multi-agent communication	Promote information sharing among agents, addressing issues such as partial observability	DIAL [107], VBC [108], I2C [109], TarMAC [110], MAIC [111], MASIA [112]	Effectively enhance cooperative capabilities in tasks with partial observability
Agent modeling	Develop techniques that enable agents to infer the actions, goals, and beliefs of other agents (entities) in the environment	ToMnet [113], OMDDPG [114], LIAM [115], LILI [116], MBOM [117], MACC [118]	Mitigate the issue of environmental non-stationarity caused by the presence of other agents
Model-based MARL	Leverage offline or sampled data to learn an environment model, reducing the number of interactions, and improving sample efficiency	MAMBPO [119], AORPO [120], MBVD [121], MAMBA [122], VDFD [123]	Utilize successful model learning methods or develop approaches tailored to MAS to improve sample efficiency in complex scenarios
Action hierarchical learning	Decompose complex problems by learning separately under sub-tasks	FHM [124], HSD [125], RODE [126], ALMA [127], HAVEN [128], ODIS [34]	Enhance the cooperation efficiency of multi-agent systems in complex scenarios
Topological structure learning	Utilize coordination graphs to model interactions between individuals in MAS	CG [129], DCG [130], DICG [131], MAGIC [132], ATOC [133], CASEC [134]	Explicitly (or implicitly) characterize the relationships between agents to reduce the joint action space of the system
Other aspects	Conduct research in areas such as interpretability, theoretical analysis, social dilemmas, large-scale systems, and delayed rewards	NA2Q [135], ACE [136], CM3 [137], MAHHQN [138], Refs. [139∼142]	Further complete the research on cooperative MARL

虑智能体间的交互以设计基于协作探索的方法, 在多类环境上表现出较好的协作性能. CMAE [104] 设计受限空间选择来鼓励多智能体探索价值更高的区域以提升协作效率. EMC [103] 将单智能体中基于动力的探索方法扩展到多智能体领域并且提出基于情景记忆的方法以存储高价值轨迹进而促进探索.

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UneVEn [105] 通过同时学习多组任务以得到一个多智能体通用继承性特征来提升智能体在新任务上的探索效率, 达到解决多智能体中的相对过拟合问题的目的. SMMAE [106] 通过均衡多智能体系统中的个体探索与团队探索的关系以提升协作效能, 在复杂任务场景下取得令人满意的效果. ADER [144] 最近被开发出来以研究如何平衡多智能体协作任务中的探索与利用问题.

此外, 也有部分方法从其他方面对多智能体探索进行研究, 包括多智能体内部奖赏探索 [145]、分布式场景下的多智能体探索 [146,147]、基于信息获取的探索方法 [148]、未知初始化条件下的协作探索 [149]、多智能体多臂赌博机下的探索问题 [150]、多智能体竞争场景下的探索与利用均衡 [151]、基于组成结构的探索方法 [152]、乐观探索 [153] 与奖赏稀疏场景下的探索 [154] 等. 虽然以上方法从各方面对多智能体探索进行研究, 并在一些测试环境上取得不错的效果, 但如何开发针对各类探索算法的测试环境与提出有效且完备的多智能体探索理论等都是未来值得研究的课题与方向.

3.3.2　多智能体通信

通信在多智能体系统中扮演着重要的角色, 系统内部的智能体基于通信技术分享经验、意图、观测等重要信息 (如图 10 所示), 获得对环境和对其他智能体更清晰的认知, 最终更好地协作并完成相应任务 [20]. 智能体通信一般涉及 3 个要素, 在何时与哪个 (些) 智能体交换何种信息. 早期方法的主要研究内容是如何更好地将通信与现有的多智能体强化学习相结合以提高多智能体协作效率, 其中 DIAL [107] 是一种简单的通信机制, 其智能体向所有队友广播信息, 以实现端到端的强化学习训练. 然而, 该方法仅限于离散信息通信场景. CommNet [155] 提出一种高效的集中式通信结构, 其中所有智能体的隐藏层输出被收集并平均以增强局部观测. 这一方法由于存在信息语义丢失, 限制了其在复杂场景中的性能. VBC [108] 添加一个基于方差的正则化器来消除信息中的噪声成分, 实现了较低的通信开销和比其他通信方法更好的性能. 然而, VBC 需要将智能体的局部信息的隐变量发送给所有队友, 不可避免地导致带宽浪费. 为了减轻由于信息冗余导致的本地策略负担, IC3Net [156], Gated-ACML [157] 和 I2C [109] 等研究工作学习一个门控机制来决定在最适当的时机与最合适的队友进行通信. SMS [158] 将多智能体通信建模为一个合作博弈问题, 通过夏普利值 (Shapley value) 评估每条信息对于决策的重要性, 裁剪没有正增益的信道. MASIA [112] 引入信息聚合的概念, 通过自监督信息模块在信息接收端有效提取信息进行决策; 另一方面, 该文还首次设计开源的离线通信数据集, 在在线和离线场景下多种实验表明所设计的通信结构的高效性与合理性.

也有部分工作对通信内容的确定展开研究. TarMAC [110] 在信息发送端通过附加签名产生信息, 智能体在信息接收端使用注意力机制计算其局部观测与发送自不同队友的信息签名的相似度, 得到一个有针对性的综合信息, 并将该综合信息用于增强智能体的局部策略空间. 然而, TarMAC 需要将智能体的信息广播给所有队友, 导致了严重的带宽浪费. DAACMP [159] 基于 MADDPG [28] 框架, 在其中添加一个双重注意力机制, 表明注意力可以提高多智能体系统的协作性能. 然而, DAACMP 也缺乏对队友的感知, 无法实现定制化的通信目的. NDQ [160] 利用最小化通信范式来设计两个不同信息论正则化项, 从而优化值函数, 实现局部可观测场景下的近似值分解算法, 在一些需要强通信的协作任务中取得较好协作性能. NDQ 正则化的信息会增大本地策略空间, 损害学习过程, 并且信息的不确定性限制其在复杂环境中的性能. TMC [161] 应用平滑和动作选择正则化器实现简洁而稳健的通信, 将收到的信息作为激励添加到个体值函数中, 这是一种高效且有前景的方法. 然而, TMC 缺乏队友建模, 同时采用了广播通信方式, 妨碍其在复杂场景中的协作性能. MAIC [111] 设计一种基于队友建模的分布式激励性通信, 在多类任务场景中取得了优异的通信效能, 最近也有部分工作, 如 CroMAC [162], AME [163], $\mathcal{R}$-MACRL [49] 与 MA3C [164] 等, 关注通信策略在部署过程中的健壮性, 以获得在策略部署过程中可能面临的通信干扰问题. 虽然当前方法在多种任务场景中取得了一定的效果, 当前多智能体环境与人类环境存在天然的鸿沟, 也就是多智能体所处世界与人类社会世界并不相通, 如何通过使用如大语言模型技术等打通两者的鸿沟 [165], 促进人机交互, 以及对通信内容进行可解释性研究及理论

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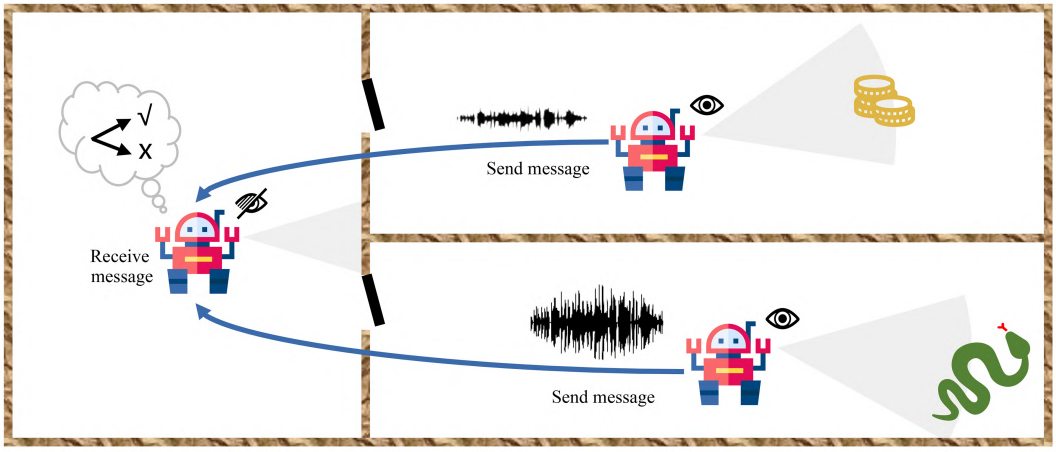

图 10　(网络版彩图) 多智能体通信示意图.

Figure 10　(Color online) Diagram of multi-agent communication.

分析等都是未来值得研究的内容.

3.3.3　基于智能体建模的协作方法

赋予智能体以推断环境中的其他智能体的动作、目标和信念是多智能体强化学习尤其是协作任务中的重点之一 [59], 通过有效且准确的建模技术, 智能体可以高效协作 (如图 11 所示). 一种最直接的方式是使用心理学中的心智理论 (theory of mind, TOM) 到多智能体系统中以促进智能体协作, 其中 ToMnet [113] 及后续的部分工作, 如 ToM2C [166], CTH [167], 文献 [168] 等, 将人类心理学理论引入到多智能体任务中, 在一些任务中起到队友建模的作用, 可以提升多智能体间的协作能力, 更多关于 TOM 的进展可以参见文献 [169].

另一方面, 其他部分工作通过其他技术去尝试对其他队友进行建模, OMDDPG [114] 运用变分自编码器 (variational autoencoders, VAEs), 借助智能体本地信息去推断其他智能体的行为; 进一步, LIAM [115] 将该技术扩展到局部可观测条件下, 并开发一种高效的队友建模技术. LINDA 通过对队友感知以缓解局部可观测问题, 在多种任务场景下协助提升多智能体系统的协作能力 [170]. MACC [118] 利用局部信息学习子任务表征并用于增广智能体策略. MAIC [111] 通过局部观测信息预测队友行为并生成激励性消息以促进协作. LILI [116] 考虑机器人场景下的智能体建模问题, 开发基于智能体高层策略表征学习的算法以促进协作. SILI [171] 在 LILI 的基础上进一步考虑学习一个平滑的表征以建模其他智能体. 文献 [172] 赋予机器人理解自己的行为, 并开发技术以更好地帮助人类, 达到快速价值对齐的目的.

除了前述所涉及的建模技术之外, SOM [173] 让智能体学习通过使用自己的策略去预测队友的行为, 并基于此在线更新自己对于其他队友的未知的目标信念. MBOM [117] 开发基于模型学习的对手建模技术. 文献 [174] 在智能体策略学习的同时, 将对其他智能体的行为建模作为附加的目标以优化策略学习. DPIQN 与 DRPIQN [175] 考虑建模有策略变化的多智能体任务场景, 提出使用策略特征优化器以优化策略学习. ATT-MADDPG [176] 通过基于注意力机制的多智能体版本 DDPG [177] 算法以得到一个联合队友表征来评估队友的策略, 以促进多智能体协作. 文献 [178] 及其后续工作考虑通过诸如概率推断技术等解决队友建模中的循环推断问题. 文献 [179] 提出 TeamReg 与 CoachReg 以相应评估合作团队的动作选择以及更好地进行队友协作. 前述工作在多种任务场景下表现出不错的协作效果, 然而在这些工作中, 智能体需要对环境中的其他智能体或者实体进行建模, 当处理大规模智能体系统或实体较多时, 使用上述方法会极大增加计算复杂度, 如何开发诸如分布式分组等技术手段等以提升其计算效率等, 是未来的研究方向之一.

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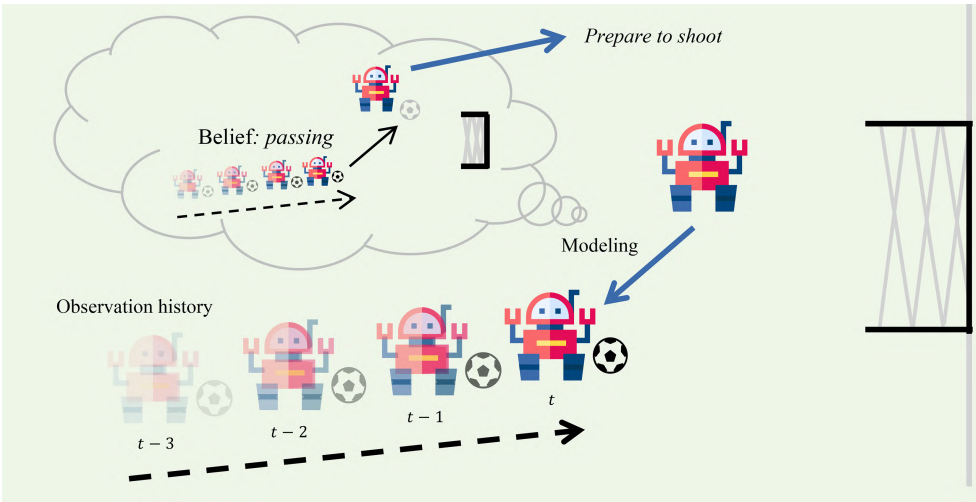

图 11　(网络版彩图) 多智能体建模示意图.

Figure 11　(Color online) Diagram of multi-agent modeling.

3.3.4　基于模型的多智能体协作

强化学习由于需要与环境持续交互以进行策略优化, 大部分算法是无模型的 (model-free), 即智能体仅使用与环境交互的样本用于策略优化, 往往存在样本利用率低下的问题. 基于模型的强化学习是一种被认为具有较高样本利用率的方法, 一般的基于模型的方法通常需要学习环境的状态转移函数 (通常也包括奖赏函数), 即环境模型, 然后基于环境模型的知识去进行策略优化 [180, 181], 可以显著提升样本效率.

近些年来, 基于模型的多智能体强化学习也逐步得到关注并取得一定的进展 [35]. 早期的一些工作尝试将单智能体中成功应用的技术扩展到多智能体场景, MAMBPO [119] 将单智能体中成功应用的方法 MBPO [182] 扩展到多智能体任务场景, 构建基于 CTDE 范式与世界模型的多智能体强化学习, 可以一定程度提升系统的采样效率. CTRL [183] 与 AORPO [120] 进一步将对手建模技术引入到模型学习中以获取自适应的轨迹数据以增广原始数据. MARCO [184] 学习一个多智能体世界模型并用其学习一个集中式的探索策略以在高不确定区域收集更多增广数据以学习多智能体策略. 也有部分技术将在单智能体中广泛应用的 Dreamer [185] 技术扩展到多智能体任务中, MBVD [121] 学习一个基于值分解的隐式的世界模型, 在该世界模型中, 智能体可以在隐空间中对状态价值进行评估, 以使得其拥有 “先见之明”. MAMBA [122] 在合作任务场景中通过基于模型的强化学习来进一步加强集中式训练过程, 该文发现在训练过程中可以通过世界模型进行算法优化, 在执行过程中可以通过通信搭建一个新的相应的世界模型. VDFD [123] 则通过开发学习解耦世界模型的技术, 极大提高多智能体强化学习的样本利用率.

另外一方面, 研究者也从其他角度探索多智能体系统中世界模型的学习, 比如分布式机器人控制中的防撞 [186]、安全效率提升 [187]、智能体交互建模 [188]、分布式网络系统优化 [189]、基于模型的对手建模 [117]、基于模型学习的多智能体通信效率提升 [190~192]、基于模型的平均场多智能体强化学习 [193]、高效多智能体模型学习 [194]、模型学习以提升离线多智能体强化学习的学习效率 [195] 与基于模型多智能体强化学习的应用 [196] 等. 虽然以上方法在一些方面取得一定的成果, 但是随着智能体数量变多带来的维度爆炸问题以及分布式执行带来的部分可观测问题, 往往阻碍基于模型的多智能体强化学习算法的发展, 如何开发可以针对多智能体特色的高效的方法, 利用诸如子集选择等手段 [197] 是后续值得研究的课题与方向之一.

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3.3.5　多智能体分层强化学习与技能学习

强化学习在复杂场景可能面临维度灾难问题, 为了解决维度灾难, 研究者提出了分层强化学习 (hierarchical reinforcement learning, HRL) [198]. HRL 的主要目标是将复杂的问题分解成多个小问题, 分别解决小问题从而达到解决原问题的目的. 近年来, 多智能体分层强化学习也取得一些进展 [199], FHM [124] 将封建领主结构引入多智能体强化学习过程中, 该方法主要针对智能体有不同任务目标的场景, 无法适用于共有目标下的协作任务场景. 为了解决多智能体任务中的稀疏与奖赏延迟问题, 文献 [200] 提出分层版本的 QMIX 算法与分层通信结构, 该方法在多个任务场景下取得优异的协作效果, 然而该方法需要手动设计高层动作. HSD [125] 通过一个上层宏策略输出相应的技能, 该方法的整体结构主要通过监督学习进行训练. RODE [126] 学习动作表征, 并通过分层的方式选择动作以提升算法在多种场景下的协作性能. VAST [201] 通过分层解决大规模多智能体协作任务中的效率低下问题. ALMA [127] 充分发掘多智能体任务中的任务结构, 并相应提出上层子任务分解策略与下层智能体行动策略. HAVEN [128] 提出一种智能体间和智能体内的双向分层结构以进一步提升多智能体的协作效率.

除前述文章之外, 在单智能体强化学习中部分基于技能的工作取得一定进展 [202], 也在多智能体强化学习中得到应用. 文献 [203] 考虑如何将技能整合到多智能体系统中以提升系统性能, HSL [204] 设计方法从多种任务中提取出异质且足够多样的多智能体技能以供下游任务学习, ODIS [34] 则关注于从离线数据集中提取技能, MASD [205] 通过最大化潜在技能分布与所有智能体状态与技能组合之间的互信息, 期望发现有益于协作的技能. SPC [206] 关注于多智能体强化学习中的技能种群的自动课程学习过程, 文献 [207] 也关注于如何通过学习扩充多智能体技能等. 前述所讲方法在一定程度上学习或者使用了多智能体中的技能, 然而技能的表现目前仍然缺乏可解释性, 如何通过诸如自然语言打通多智能体与人类世界的边界 [208], 使得学到的技能具备可解释性, 是后续值得研究的方向与话题之一.

3.3.6　协作多智能体拓扑结构学习

智能体间的交互是多智能体问题研究中的重点之一, 协作图是一种可以通过将多智能体值函数以图的方式进行分解以显式刻画智能体间的交互关系的方法 [129, 209], 最近几年在协作多智能体强化学习中得到广泛应用. 在基于协作图的多智能体强化学习任务中, 一般可以用节点表示智能体, 边 (超边) 表示有连接关系的智能体间构建在联合动作观测空间上的收益函数, 一个协作图可以表示多智能体间价值分解的高阶形式. 一般可以通过利用如分布式受限优化 (distributed constraint optimization, DCOP) 算法 [210] 等寻找具有最大值的动作选择, 智能体可以通过连接的边进行多轮信息交换. DCG [130] 引入深度学习的一些技巧将协作图扩充到高维状态动作空间中, 在复杂的任务场景如 SMAC 上可以改善多智能体间的相对过泛化问题, DCG 主要专注于在提前训练好的静态和稠密拓扑结构上, 在动态环境中可扩展性较差, 需要稠密且低效的通信方式.

在协作图问题中的核心问题是如何学到一个动态且稀疏, 可以满足智能体动作选择的图结构, 该类问题在多智能体学习领域一直得到广泛关注. 其中稀疏协作 Q- 函数学习 [211] 尝试为价值函数学习稀疏的图结构, 该方法学到的图结构是静态的并且需要大量的先验知识. 文献 [212] 提出为每个智能体学习最小化的动态图集合, 该方法存在的问题是计算复杂度随着智能体邻居数量的增多呈指数上升. 文献 [213] 通过研究一些图结构的表达能力, 但是主要专注于随机拓扑结构与无状态问题. 文献 [134] 提出新的协作图测试环境 MACO, 并且提出基于上下文感知的稀疏协作图算法 CASEC, 该算法通过学习导致表征以有效减少由于图构建过程中带来的收益函数评估误差, 并且在多个环境上取得较好效果. 后续工作也从多方面提升协作图, 如开发非线性协作图结构 [214]、自组织多项式协作图结构 [215] 等.

另一方面, 一些方法也通过利用诸如注意力机制等技术去寻找多智能体间的隐式图结构, 通过借助注意力机制去剪断部分不需要的连接结构. 如 ATOC [133] 通过注意力机制学习最必要的通信结构,

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DICG [131] 通过注意力机制学习隐式的协作图结构, MAGIC [132] 以及通过图注意力机制提升多智能体系统的通信效率与组队. 以上方法虽然可以在一定程度上得到或者改善多智能体系统交互之间的拓扑结构, 但是目前这些方法一般仅能在智能体较少的规模下取得进展, 如何在大规模且存在强交互的场景下得到最优拓扑结构, 是未来值得研究的方向之一 [216].

3.3.7　其他方面

除了前文所介绍的目前研究较充分的课题之外, 包括多智能体策略模仿 [217~219]、多智能体协作可解释性 [135]、开发具备决策先后顺序的多智能体协作算法 [32, 136, 220, 221]、多智能体协作理论分析 [22, 222~224]、多智能体多任务多阶段协作学习 [137]、多智能体系统社会困境问题 [225]、异步多智能体协作 [226, 227]、大规模多智能体协作 [228]、延时奖赏下的多智能体协作 [229]、混合动作空间下的多智能体协作 [138]、多智能体因果关系发现 [63]、多智能体协作中的课程学习 [139]、协作多智能体强化学习中的教师与学生关系研究 [140]、多智能体中的公平性研究 [141] 与基于实体关系的多智能体协作 [142] 等在内的课题, 目前也逐步被探索与挖掘.

3.4　典型测试场景

在算法设计研究的同时, 目前有部分工作开发了一系列测试环境对算法进行多方面的验证, 典型环境包括星际争霸多智能体挑战 (starcraft multi-agent challenge, SMAC) [33] 及其改进版本 SMACv2 [230] 和 SMAClite [231]、多智能体粒子世界 (multi-agent particle world, MPE) [28]、多智能体版本的 MuJoCo (multi-agent MuJoCo, MAMuJoCo) [97]、谷歌足球 (Google research football, GRF) [232]、大规模多智能体验证环境 MAgent [233]、多智能体离线强化学习测试环境 [234] 与多智能体出租车环境 TaxAI [235] 等. 常见多智能体测试环境及其特点如表 2 [26~28, 33, 88, 97, 112, 134, 155, 160, 225, 228, 232~234, 236~257] 所示. 另外, 为了后续工作的方便, 也有一些研究者将当前主流的测试环境进行集成并开源, 主要包括 Pymarl [33]2), EPyMARL [236]3), PettingZoo [258]4), MARLlib [259]5) 与 “及第”6) 等.

3.5　典型应用案例

与此同时, 协作多智能体强化学习算法也在多类任务场景中得到广泛应用, 包括在游戏领域、工业应用、机器人控制、交叉领域以及国防领域等 [17, 260~263].

早期的多智能体算法主要聚焦于游戏领域, 通过应用强化学习算法以优化策略, AlphaStar [6] 以强化学习为基础, 应用诸如种群训练等多智能体学习算法, 在星际争霸游戏中控制己方智能体击败敌方, 获得了惊人的表现 [16, 264]. 后续很多工作也将多智能体强化学习算法应用在其他游戏任务中, ViVO 团队通过分层强化学习控制王者荣耀中英雄单位进行游戏, 在多个场景中取得了良好的对抗效果 [265], 包括 SMAC [33] 在内的诸多测试环境都是基于游戏引擎所开发的. 另一方面, 除了实时游戏之外, 多智能体强化学习在如西洋陆军棋 [266]、中国象棋 [267]、麻将 [268]、斗地主 [269]、足球 [232]、篮球游戏 [244, 270] 与捉迷藏 [243] 等方面也都得到人们的关注并且在多种任务场景下取得优异的成果.

另一方面, 一些研究者也尝试将多智能体强化学习应用到工业领域, 这些研究主要借助多智能体强化学习在问题求解方面的巨大潜力, 将工业问题建模为多智能体强化学习任务. 比如文献 [271] 通过将无人驾驶建模为多智能体协作问题, 并使用基于协作图的方式促进各无人车之间的协作, 发现在多种任务场景下可以取得自动实现车辆调度, 其他相关工作也通过多智能体强化学习提升交通信号控制 [272]、无人驾驶 [273]、无人机控制 [274, 275]、无人车控制 [276] 等. 也有部分研究者从其他方面展开研

2) https://github.com/oxwhirl/pymarl.
3) https://github.com/uoe-agents/epymarl.
4) https://github.com/Farama-Foundation/PettingZoo.
5) https://marllib.readthedocs.io/en/latest/index.html.
6) http://www.jidiai.cn/environment.

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表 2　典型多智能体测试环境介绍.

Table 2　Introduction to classic multi-agent benchmarks.

Environment	Heterogeneous agent	Scenario type	Observation space	Action space	Typical number	Communication capability	Problem domain
Matrix Games [88] (1998)	Yes	Mixed	Discrete	Discrete	2	No	Matrix games
MPE [28] (2017)	Yes	Mixed	Continuous	Discrete	2∼6	Allowed	Particle games
MACO [134] (2022)	No	Mixed	Discrete	Discrete	5∼15	Allowed	Particle games
GoBigger [237] (2022)	No	Mixed	Continuous	Continuous or discrete	4∼24	No	Particle games
MAgent [233] (2018)	Yes	Mixed	Continuous+image	Discrete	1000	No	Large-scale particle confrontation
MARLÖ [238] (2018)	No	Mixed	Continuous+image	Discrete	2∼8	No	Adversarial games
DCA [228] (2022)	No	Mixed	Continuous	Discrete	100∼300	No	Adversarial games
Pommerman [239] (2018)	No	Mixed	Discrete	Discrete	4	Yes	Bomberman game
SMAC [33] (2019)	Yes	Cooperative	Continuous	Discrete	2∼27	No	StarCraft game
Hanabi [240] (2019)	No	Cooperative	Discrete	Discrete	2∼5	Yes	Card game
Overcooked [241] (2019)	Yes	Cooperative	Discrete	Discrete	2	No	Cooking game
Neural MMO [242] (2019)	No	Mixed	Continuous	Discrete	1∼1024	No	Multiplayer game
Hide-and-Seek [243] (2019)	Yes	Mixed	Continuous	Discrete	2∼6	No	Hide and seek game
LBF [236] (2020)	No	Cooperative	Discrete	Discrete	2∼4	No	Food search game
Hallway [160] (2020)	No	Cooperative	Discrete	Discrete	2	Yes	Communication corridor game
GRF [232] (2019)	No	Cooperative	Continuous	Discrete	1∼3	No	Soccer confrontation
Fever Basketball [244] (2020)	Yes	Mixed	Continuous	Discrete	2∼6	No	Basketball confrontation
SUMO [245] (2010)	No	Mixed	Continuous	Discrete	2∼6	No	Traffic control
Traffic Junction [155] (2016)	No	Cooperative	Discrete	Discrete	2∼10	Yes	Communication traffic scheduling
CityFlow [246] (2019)	No	Cooperative	Continuous	Discrete	1∼50+	No	Traffic control
MAPF [247] (2019)	Yes	Cooperative	Discrete	Discrete	2∼118	No	Path navigation
Flatland [248] (2020)	No	Cooperative	Continuous	Discrete	>100	No	Train scheduling
SMARTS [249] (2020)	Yes	Mixed	Continuous+image	Continuous or discrete	3∼5	No	Autonomous driving
MetaDrive [250] (2021)	No	Mixed	Continuous	Continuous	20∼40	No	Autonomous driving
MATE [251] (2022)	Yes	Mixed	Continuous	Continuous or discrete	2∼100+	Yes	Target tracking
MARBLER [252] (2023)	Yes	Mixed	Continuous	Discrete	4∼6	Allowed	Traffic control
RWARE [236] (236)	No	Cooperative	Discrete	Discrete	2∼4	No	Warehouse logistics
MABIM [253] (2023)	No	Mixed	Continuous	Continuous or discrete	500∼2000	No	Inventory management
MaMo [27] (2022)	Yes	Cooperative	Continuous	Continuous	2∼4	No	Parameter tuning
Active Voltage Control [26] (2021)	Yes	Cooperative	Continuous	Continuous	6∼38	No	Power control
MAMuJoCo [97] (2020)	Yes	Cooperative	Continuous	Continuous	2∼6	No	Robot control
Light Aircraft Game [254] (2022)	No	Mixed	Continuous	Discrete	1∼2	No	Intelligent air combat
MaCA [255] (2020)	Yes	Mixed	Image	Discrete	2	No	Intelligent air combat
Gathering [225] (2020)	No	Cooperative	Image	Discrete	2	No	Social dilemma
Harvest [256] (2017)	No	Mixed	Image	Discrete	3∼6	No	Social dilemma
Safe MAMuJoCo [257] (2023)	Yes	Cooperative	Continuous	Continuous	2∼8	No	Safe multi-agent
Safe MARobosuite [257] (2023)	Yes	Cooperative	Continuous	Continuous	2∼8	No	Safe multi-agent
Safe MAIG [257] (2023)	Yes	Cooperative	Continuous	Continuous	2∼12	No	Safe multi-agent
OG-MARL [234] (2023)	Yes	Mixed	Continuous	Continuous or discrete	2∼27	No	Offline dataset
MASIA [112] (2023)	Yes	Cooperative	Discrete or continuous	Discrete	2∼11	No	Offline communication dataset

究, 比如通过多智能体强化学习进行电力控制, 其中文献 [26] 将主动电力控制建模为 Dec-POMDP 问题, 对环境进行开源, 并在该环境下对多种多智能体算法进行验证. 一些工作也从能源方面展开研究, 比如通过多智能体强化学习算法控制风力发电的频率 [277]; 文献 [278, 279] 在分布式马尔可夫决策过程建模下, 分别提出利用基本 MADDPG 与结合平均场近似的方法针对能源交易双边拍卖问题进行求解. 也有部分工作在金融领域使用多智能体强化学习 [280∼282]. 除此之外, MAAB [283] 设计一种基于多智能体强化学习的在线自动竞价框架, 以及其他的问题 [284]. 有工作尝试通过多智能体强化学习进行 FPGA 时钟同步 [285]. 虚拟淘宝 [286] 利用多智能体对抗模仿学习捕捉用户偏好. 也有部分工作通过多

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智能体强化学习对机器人进行控制 [287~289], 如文献 [290] 通过将协作多智能体强化学习应用到机器人手术的合作协助任务中, 表明可以显著提升其任务完成度.

多智能体强化学习也在一些交叉领域得到应用. 比如 MA-DAC [27] 将多参数优化问题建模为多智能体强化学习任务, 并通过协作多智能体强化学习算法如 QMIX 解决该问题, 发现基于多智能体强化学习方法可以显著提升多参数优化能力. MA2ML [291] 通过多智能体强化学习有效处理自动化机器中模块的连接优化学习问题. 文献 [292] 研究丰富视觉信息条件下的多智能体强化学习, 并设计协作场景下的导航任务. 文献 [293] 提出一种基于协作对齐的多摄像头协作系统以解决主动多目标跟踪问题, 具体而言, 将每一个摄像头当成一个智能体, 使用任意一种多智能体算法对其进行控制. 文献 [294] 将图像数据增强建模为多智能体问题, 该文提出一个更细粒度的自动化数据方法, 把一幅图像划分为多个网格, 并寻找联合最优增强策略. 除此之外, 不少工作关注于将多智能体强化学习算法应用到组合优化中, 如文献 [295] 研究通过多智能体强化学习优化在线泊车分配, 文献 [296] 专注于通过多智能体强化学习框架研究锂离子电池调度问题, 文献 [297] 利用多智能体强化学习解决资源抢占环境下的作业调度问题, 文献 [298] 探索通过多智能体强化学习研究小型工厂中的在线调度问题. MARLYC [299] 通过提出一种称为多智能体强化学习偏航控制的新方法以建议控制每台涡轮机的偏航, 最终提高农场的总发电量. 协作多智能体强化学习在日常生活中也得到一些应用, 文献 [300] 通过设计两个协作的机器人对房间内的声学效果进行测试, 文献 [301] 通过多智能体系统对日常家居中的能量进行管理, Botzone [302] 研究基于多智能体强化学习的在线智能教育, 也有部分工作尝试将多智能体协作技术应用到医疗领域, 比如通过多智能体协作对神经元 [303] 或医疗图像进行分割 [304] 等.

除了前文所介绍的应用, 多智能体强化学习也在国防应用领域得到一定的探索 [305~310]. 文献 [311] 基于 MADDPG 与注意力机制提出一种可以处理队友可变与对手可变场景的多智能体空战任务场景. 文献 [312] 提出一种分层多智能体强化学习 (hierarchical multi-agent reinforcement learning, HMARL) 方法解决异构无人机群协同决策问题, 主要针对典型的压制敌方防空 (suppression of enemy air defenses, SEAD) 任务, 将其解耦为两个子问题, 即高层目标分配 (target allocation, TA) 子问题和低层协同攻击 (cooperative attacking, CA) 子问题. 文献 [313] 搭建一种基于多智能体强化学习的多无人机协作空战系统, 仿真结果表明该文设计的策略学习方法可以获得较高的能量优势并且有效击败多种对手. 文献 [314] 提出一种用于空对空作战的分层多智能体强化学习框架以处理多个异构智能体场景. MAHPG [315] 设计了一种基于自博弈对抗训练与分层决策网络的策略梯度多智能体强化学习方法以提升系统空战性能, 达到可以学习多种策略的目的. 此外, 采用层次决策网络来处理复杂的混合动作文献 [316] 设计一种基于两阶段图注意神经网络的机制以捕捉智能空战场景下的智能体间的关键交互关系, 实验表明该方法可以在大规模空战场景显著提升系统的协作能力.

4　开放环境下的协作策略学习

前述内容主要是在经典封闭环境, 也就是环境中的要素恒定不变的条件下进行讨论的, 真实环境中的机器学习算法研究往往需要面对某些要素发生变化的情况, 这一特点催生了新的研究领域 – 开放环境下的机器学习, 包括 Open-world Learning [317], Open-environment Learning [38], Open-ended Learning [318], Open-set Learning [319] 等.

传统机器学习一般在经典封闭环境下进行讨论, 在这类任务场景下, 环境中的重要因素不会发生改变, 这里的重要因素在不同的研究领域可以有不同的定义和范围, 比如在监督学习中可以是新出现的类别、持续学习的新任务、神经网络输入中的特征增/减、训练 (测试) 数据分布的变化、任务间学习目标的变化等. 根据不同的理解和称谓, 开放环境下的机器学习 (open-environment machine learning) [38] 一定程度上与开放学习 (open-ended learning) 存在联系 [318], 与封闭环境下的机器学习相对应, 开放环境下的机器学习主要考虑机器学习环境中的重要因素可能发生改变的情况 [38]. 过去在该方面有部分

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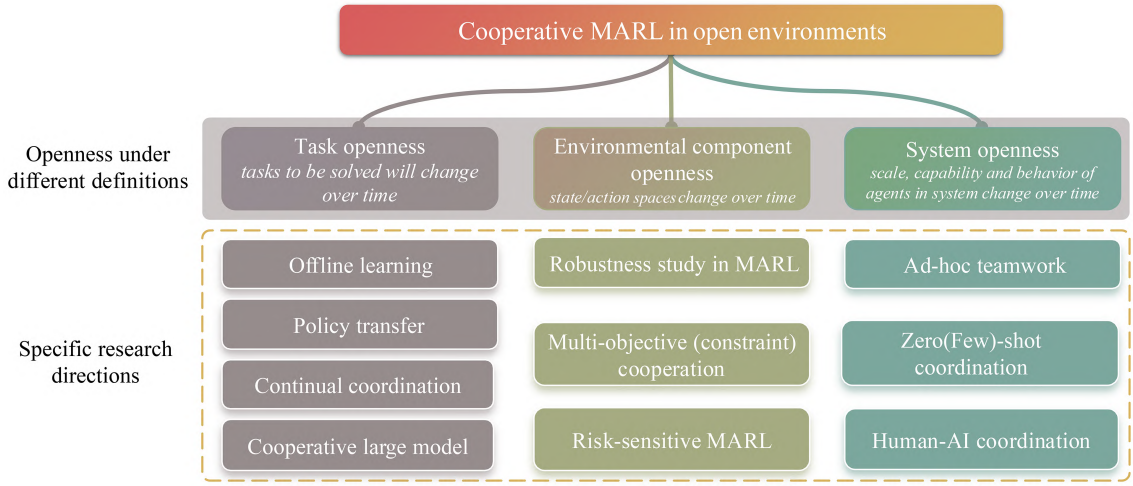

图 12　(网络版彩图) 开放环境下的协作多智能体强化学习分类框架图.

Figure 12　(Color online) Classification framework of cooperative multi-agent reinforcement learning in open environments.

工作展开研究, 包括类别变化监督学习中的类别增减 [37]、特征变化 [320], 持续学习中的任务变化 [64], 开放环境下的测试环境设置 [321], 博弈问题中的开放性研究 [322] 等.

另外一方面, 开放环境下的强化学习近年来也引起不少关注并且在不同方面取得一些进展, 不同于监督学习从提供的有标记训练数据中进行学习, 或无监督学习对给定的数据内在信息进行分析, 强化学习中的智能体被置于一个陌生的环境, 需要自主与环境交互, 并从交互的结果数据中学习. 不同的交互方式会产生不同的数据, 因此强化学习中的一个关键难点是从动态的数据分布中学习, 而学习的结果还会进一步改变数据的分布. 另外, 强化学习中的智能体需要考虑所处环境 MDP 的变化, 过去的一些方法尝试学习一个具备可信性的强化学习策略 [39]、通过诸如演化学习等方式学习一个具备高泛化能力的策略 [40]、学习技能以面向开放环境 [323]、通过改变奖赏函数以增强泛化能力 [324]、面向开放强化学习进行测试环境设计 [325]、基于强化学习设计通用开放智能体 [326,327]、提升策略泛化能力 [42] 等.

4.1　开放环境下的协作多智能体强化学习

前述内容介绍了部分开放环境下的机器学习与单智能体强化学习相关工作, 也有部分研究考虑了多智能体系统的开放性, 即某些智能体在协作过程中会加入或离开团队 [328]. 由于经典的方法假设系统规模与组成因素保持恒定不变, 智能体无法及时感知到队友行为风格的变化, 对此, 在多智能体规划方面, 有相关工作通过建立开放式分布式部分可观测马尔可夫决策过程 (open dec-POMDP) [329]、团队 POMDP (team-POMDP) [330,331]、轻量级个体 POMDP (I-POMDP-Lite) [332,333] 和 CI-POMDP [334] 等以解决系统开放问题. 在多智能体强化学习方面, GPL [335] 将开放式即时团队合作问题 (open ad-hoc teamwork) 形式化为 OSBG (open stochastic Bayesian games), 并假设全局可观测性以提高效率, 但其在真实世界中难以实现. 此外, 它使用了基于图神经网络的方法, 仅适用于单个可控智能体的设定, 导致其扩展到多个可控智能体时比较困难, 文献 [336] 最近提出开放多智能系统 OASYS, 以对开放多智能体系统分类进行描述.

虽然前述工作对开放环境下的多智能体强化学习开展了一定的研究, 但其所关注的内容都相对局部且片面, 缺乏对整个研究领域的全面综述. 如图 12 所示, 基于此, 本文旨在从以下方面对协作多智能体强化学习中的开放性进行定义与分类, 并进一步划分为 10 个具体课题:

• 任务开放性: 部署在开放环境中的多智能体系统所需要解决的任务会随着时间发生变化, 存在着探索代价大、策略适应慢等问题. 针对此, 多智能体策略应具备离线策略学习、迁移、持续学习的能

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表 3　开放环境下的协作多智能体强化学习研究内容.

Table 3　Research content on cooperative multi-agent reinforcement learning in open environments.

Research direction	Core content	Representative algorithms	Applications and achievements
Offline learning	Extend the single-agent offline learning techniques to multi-agent scenarios or design targeted offline MARL algorithms	ICQ [337], MABCQ [338], SIT [339], ODIS [34], MADT [340], OMAC [341], CFCQL [342]	Learn policies from large-scale and diverse offline data to avoid issues associated with interactions
Policy transfer	Achieve transfer of policies across tasks to enable knowledge reuse	LeCTR [343], MAPTF [344], EPC [345], Ref. [346], MATTAR [347]	Facilitate knowledge reuse between tasks to improve learning efficiency
Continual coordination	Cooperative task learning in scenarios where tasks appear sequentially	Ref. [348], MACPro [349]	Extend single-agent techniques to handle cooperative task emergence in multi-agent settings
Cooperative large model	Develop large cooperative decision-making models, or utilize current large model technologies to enhance multi-agent coordination	MADT [340], MAT [32], MAGENTA [350], MADiff [351], ProAgent [212], SAMA [352]	Achieve generality through the learning of large models; additionally, leverage large language models to enhance the system's cooperative capabilities
Robustness in MARL	Learn robust policies to cope with environmental noise, teammate changes, etc.	R-MADDPG [47], Ref. [46], RAMAO [353], ROMANCE [354], MA3C [162], CroMAC [164]	Maintain robust cooperative capabilities in conditions where environmental components are subjected to noise or even malicious attacks
Multi-objective (constrained) cooperation	Optimize problems with multiple objectives and consider the optimal solutions of different objective functions	MACPO(MAPPO-Lagrangian) [257], CAMA [355], MDBC [356], Refs. [187, 357, 358]	Consider multiple constraint objectives present in the environment, making progress in safety-critical domains
Risk-sensitive MARL	Model the numerical values of variables in the environment as distributions using value distributions, and use risk functions to assess system risks	DFAC [359], RMIX [360], ROE [153], DRE-MARL [361], DRIMA [362], Refs. [363, 364]	Enhance coordination performance in complex scenarios, effectively perceive risks, and assess performance in risk-sensitive scenarios
Ad-hoc teamwork	Creating an autonomous agent capable of efficiently and robustly collaborating with unknown teammates in specific tasks or situations	Refs. [332, 365], ODITS [366], OSBG [335], BRDiv [367], L-BRDiv [368], TEAMSTER [369]	Empower the autonomous agent with the ability to collaborate rapidly with unknown teammates
Zero (Few)-shot coordination	Design paradigms to enable MASs to collaborate with unseen teammates using few or zero interaction samples	FCP [370], TrajeDi [371], MAZE [372], CSP [373], LIPO [374], Macop [349], Refs. [52, 375]	Show effective zero-shot or low-shot coordination with diverse unseen teammates in benchmark environments
Human-AI coordination	Provide support for human-intelligence (human-machine) interaction, enabling better coordination between human participants and agents to accomplish specific tasks	Refs. [376], HSP [377], Latent offline RL [378], RILI [379], PECAN [380]	Achieve certain levels of coordination between human and machine in given simulation environments or real-world robot scenarios

力; 此外, 考虑到协作任务的相似性, 开发通用协作决策大模型也具备相当的价值.

• 环境要素开放性: 由于开放环境中状态空间与动态转移等环境要素发生变化, 多智能体策略应当在如观测、动作与通信等发生改变时保持稳健性; 进一步, 在满足多环境约束时, 具备应对风险的感知能力.

• 智能体系统开放性: 在多智能体系统中, 智能体的规模、能力以及行为风格都会在协作过程中发生改变, 甚至会需要与具备不同偏好的人类进行协作. 为此, 多智能体策略需要具备自组织协作能力、零 (少) 样本适应能力, 并支持人智协同.

针对上述协作多智能体强化学习所面对的环境开放性, 本文将进一步对当前研究的主要内容、存在的问题以及未来值得进一步研究的方向依照表 3 [32, 34, 46, 47, 52, 153, 162, 164, 187, 212, 257, 332, 335, 337~380] 进行分析阐述.

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4.2　任务开放性

由于多智能体系统在部署后需要解决的任务可能会随着时间而发生改变, 对于未见任务, 策略往往难以实现直接泛化. 此时, 智能体与现实世界进行交互试错重新学习所需的代价相对较大, 可以利用环境中的样本直接从静态数据进行学习. 与此同时, 系统可以利用先前任务的知识促进在新任务上的策略迁移, 提高学习效率, 并基于持续学习方法, 避免对过往知识的遗忘. 更进一步, 利用大模型技术, 可以开发出通用的协作决策模型, 对任意任务可以实现直接泛化.

4.2.1　离线协作多智能体强化学习

离线强化学习 [381, 382] 近来吸引大量研究关注, 该类研究专注于一种无需与环境交互的数据驱动训练范式 [382]. 先前的部分工作 [383] 主要关注离线学习中的分布偏移问题, 并考虑学习行为受限策略以减轻对未见数据估计的外推误差 [384~386]. 离线多智能体强化学习则是一个相对较新且具有前景的研究方向 [387], 它从静态数据集中训练多智能体协作策略. 一类离线多智能体强化学习方法尝试从带有策略约束的离线数据中学习策略, 其中 ICQ [337] 通过仅信任离线数据有效地缓解多智能体强化学习中的外推误差. MABCQ [338] 引入一个完全分布式的离线多智能体强化学习设定, 并利用值偏差和转移归一化等技术来实现高效学习. OMAR [388] 结合一阶策略梯度和零阶优化方法, 避免不协调的局部最优解. MADT [340] 利用 Transformer 对序列建模的强大能力, 将其与离线和在线多智能体强化学习任务无缝结合起来. 文献 [339] 对离线多智能体强化学习进行研究, 明确考虑智能体轨迹的多样性, 并提出一个名为 Shared Individual Trajectories (SIT) 的新框架. 文献 [389] 提出首先训练一个教师策略, 该策略有权访问每个智能体的观测、动作和奖赏. 在教师策略确定并重组了数据集中的 “好” 行为后, 创建单独的学生策略, 并将教师策略的特征与智能体之间结构关系通过知识蒸馏赋给学生策略. ODIS [34] 提出一种新颖的离线多智能体强化学习算法, 用于从多任务数据中发现协作技能. 文献 [234] 最近发布了 Off-the-Grid MARL (OG-MARL) 框架, 用于生成离线多智能体强化学习数据集和算法评估, 该框架提供了一套初始数据集和基线, 并配备一个标准化的评估协议. M3 [390] 创新性地引入多任务和多智能体离线预训练模块思想以学习更高层次可传递的策略表征. OMAC [341] 提出一种基于耦合值分解的离线多智能体强化学习算法, 将全局价值函数分解为局部和共享部件, 并保持全局状态值和 Q- 函数之间的信用分配一致性.

4.2.2　协作策略迁移

迁移学习被认为是能够帮助提高强化学习算法样本利用率的重要方法 [391], 旨在不同的任务之间进行知识重用, 加快智能体在新任务上的策略学习. 多智能体场景下的迁移学习 [60] 研究也得到广泛关注, 相比于单智能体而言, 除了考虑任务之间的知识重用之外, 还有部分研究者针对智能体之间的知识重用进行研究. 其基本想法是希望让部分智能体选择性地重用其他智能体的知识, 从而帮助多智能体系统整体上实现更好的协作. DVM [392] 将多智能体问题建模为一个多任务学习问题, 结合不同任务之间的知识, 并且使用一种值匹配机制对这些知识进行蒸馏. LeCTR [343] 在多智能体场景下进行策略教学, 让一部分智能体对其他智能体进行引导, 从而帮助整体实现更好的策略协作. MAPTF [344] 提出了一种基于选项的策略迁移方法来辅助多智能体协作.

另一方面, 多智能体任务之间的策略重用方法强调重用旧任务上的知识经验来辅助新任务的策略学习, 关注不同任务之间的知识迁移. 相比于单智能体问题而言, 不同规模的多智能体任务之间环境输入的维度可能有所不同, 这给任务之间的策略迁移带来了阻碍. DyMA-CL [393] 设计一种智能体数量无关的网络结构, 并且结合课程学习的思路提出了一系列的迁移机制来加速多智能体协作策略的学习过程. EPC [345] 提出了一种基于演化算法的多智能体课程学习方法, 来帮助群体更好地实现复杂场景下协作策略的学习. UPDeT [394] 和 PIT [395] 则借助 Transformer 网络的泛化性来处理环境输入维度变化的问题, 帮助智能体群体实现高效协作和任务之间的知识迁移. 这些多智能体迁移学习的相关工作

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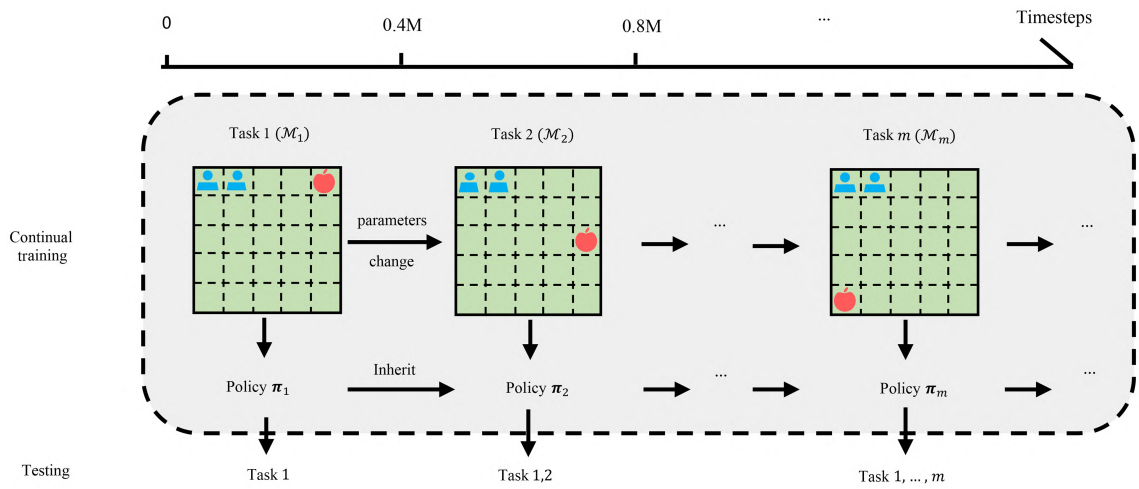

图 13　(网络版彩图) 持续协作示意图.

Figure 13　(Color online) Diagram of continual coordination.

为任务之间的知识迁移提供了一些启发, 但它们没有显式地考虑任务之间的相关性, 如何利用任务之间的相关性进行更加高效的知识迁移仍然是非常开放的研究课题. MATTAR [347] 针对协作策略模型难适应新任务的问题, 提出了基于任务关系建模的策略迁移方法. 文献 [396] 考虑使用边界迁移学习以促进多智能体强化学习.

4.2.3　多智能体强化学习与持续协作

持续学习与增量学习以及终身学习有关, 上述学习方式都假定任务或样本以顺序的方式出现 [397]. 近年来, 持续强化学习 [41,398] 受到一定关注, 在该设定下, 智能体面临的挑战是避免灾难性遗忘的同时学会将旧任务的知识迁移到新的任务 (又称稳定性 – 可塑性困境 [399]), 同时保持对大量任务的可扩展性. 研究者提出多种方法来解决这些挑战, EWC [400] 使用基于 l_2 距离的权重正则化项去约束当前模型参数和过去学习的模型参数之间的差距, 该方法需要额外的监督信息来选择特定的 Q- 函数头以及为不同的任务场景设置特定的探索方案. CLEAR [401] 是一种与任务无关的持续学习方法, 在持续学习过程中不需要任务信息. 它存储一个大型的经验回放缓冲区, 并通过从过去任务中采样数据来解决遗忘问题. 其他方法如 HyperCRL [402] 和文献 [403], 利用学习的世界模型来增强学习效率. 为了解决在任务数量众多的场景中的可扩展性问题, LLIRL [404] 将任务空间分解成子集, 并使用中餐厅过程来扩展神经网络, 使持续强化学习更加高效. OWL [405] 是一种最近提出的基于多头架构来实现的高效方法. CSP [406] 通过逐步构建策略子空间, 在一系列任务上训练强化学习智能体. 另外一类可以处理可扩展性问题的手段基于 Packnet [407] 的思想逐次将任务信息编码到神经网络中并对相关任务的网络节点数进行删减. 关于多智能体持续学习问题 (见图 13), 文献 [348] 通过在 Hanabi 基础上引入一个基于多智能体学习的测试平台来研究智能体是否可以与未知智能体协作. 然而, 它只考虑单模态任务情况. MACPro [349] 提出了一种通过渐进式任务情境化实现多智能体持续协作的方法. 它使用共享的特征提取层获得一个任务的特征, 但是使用独立的策略输出头, 每个策略输出头针对特定类别的任务进行决策. Macop [408] 赋予己方多智能体系统持续协作能力, 开发基于非相容队友演化生成与高相容多智能体协作算法对抗训练范式. 在队友在回合内与回合间发生切换的多种测试环境中, 所提出的方法可以快速捕捉队友身份, 较多种对比方法具备更强的适应能力, 在面对未知场景下也具有更强的泛化能力.

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4.2.4　协作多智能体强化学习与大模型

大模型, 尤其是大语言模型 [409], 近年来在多个领域引发广泛关注并且得到应用, 目前也有部分工作关注于通用决策大模型的探索 [410, 411] 并得到一定应用. 在单智能体强化学习任务中, GATO [412], DreamerV3 [413], DT [414] 等研究工作在许多任务场景下取得令人惊喜的效果, 这些工作都依托于现有技术如 Transformer [92] 的强大的表达能力 [415]. 另一方面, 近来的一些工作也尝试学习多智能体通用决策大模型, 其中 MADT [340] 通过提供大规模数据集来促进研究, 并将其用于在多智能体强化学习环境下探讨 DT 的应用. MAT [32] 研究一种可以有效地将 MARL 转化为单智能体问题的大模型方法, 其目标是将智能体的观察序列映射到智能体的最优行动序列, 在多个任务场景下表现出比传统方法更优越的性能效果. 为了充分利用多智能体环境中的实体信息, 文献 [350] 从实体的角度提出了多智能体多游戏实体 MAGENTA, 这是对先前的时间序列建模的一个正交研究. MADiff [351] 与 DOM2 [416] 将生成式扩散模型引入到多智能体强化学习以促进系统的协作, 在多种场景下取得最优效果. SCT [417] 通过 Transformer 模型加速多智能体在线适应. 文献 [418] 构建类人的多智能体环境, “西部世界”, 以模拟并测试大规模场景下的多智能体系统. 然而由于复杂交互等原因, 多智能体通用决策大模型目前还是一个较少涉及的领域, 如何在目前已有的多种多智能体任务场景下学到一个通用的多智能体决策大模型, 可以在多种场景下进行零样本或者少样本泛化, 或者通过微调可以获得一个在未见任务场景的快速学习的策略等, 都是极具挑战但是值得研究的方向.

4.3　环境要素开放性

由于开放环境中, 状态空间与动态转移等要素会随时间发生改变, 如受天气等影响, 传感器输入的视觉信息以及路面摩擦系数等与训练时不一致, 系统的决策也会受到多方条件的约束. 在封闭环境下训练的多智能体系统面对上述环境要素的改变可能会直接瘫痪. 为此, 在多智能体策略训练时需要考虑到策略的稳健性, 在约束变化时能相应调整策略, 并根据当前情况对决策风险进行自动识别与评估, 从而应对部署阶段环境要素的开放.

4.3.1　稳健协作多智能体强化学习

强化学习中的稳健性研究近年来得到广泛关注并且取得不少进展 [419], 稳健性的研究对象主要包括对智能体不同方面的扰动, 例如状态、奖赏、动作等. 其中一类方法引入辅助对抗攻击者, 通过对待使用策略和对手系统交替的对抗训练来实现稳健性 [420~423]. 其他方法通过在损失函数中设计适当的正则化项来提高稳健性 [424~426]. 相比于对抗训练, 这类方法可以有效提高了样本利用率. 然而, 上述方法缺乏针对噪声程度与策略执行之间的稳健性可验证性保证, 针对这一问题, 相继衍生出一些可验证的稳健性方法 [356, 427~429].

目前多智能体强化学习中关于稳健性的研究开始引起人们的关注 [45], 主要困难在于多智能体相较于单智能体需要考虑更多问题, 如智能体之间存在的复杂交互带来的非稳态 [21]、信度分配 [22]、可拓展性 [24] 等. 早期的部分工作旨在研究协作策略是否具有稳健性. 例如, 针对一个用 QMIX 算法训练好的协作策略, 文献 [430] 通过强化学习训练一个针对观测的攻击者, 之后用训练好的攻击者去攻击多智能体中的某一个智能体, 结果表明当前通过多智能体强化学习算法训练出来的协作策略对观测扰动是脆弱的. 随后文献 [45] 从多智能体系统的奖赏、状态、动作几方面对当前典型多智能体强化学习算法如 QMIX, MAPPO 进行更全面的稳健性测试, 结果进一步佐证多智能体系统在面对攻击情况下的脆弱性, 表明稳健性多智能体强化学习研究的必要性与迫切性. 在多智能体强化学习稳健性提升方面, 近来也有部分工作取得一些进展, 其中部分工作关注于学习稳健合作策略来避免过拟合至特定的队友 [46] 与对手 [431]. 类似于单智能体稳健强化学习从 MDP 本身包括状态、奖赏、动作方面进行研究, R-MADDPG [47] 关注于多智能体系统的模型不确定性, 构建模型不确定下的稳健纳什均衡的概念, 在多个环境下取得最优的稳健效果. 针对多智能体系统中部分智能体动作被扰动的问题, 文献 [48] 引入

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一个启发式的规则以及相关均衡理论来学习一个稳健的多智能体协作策略. 多智能体通信的稳健性也在最近几年得到了一些关注. 文献 [432] 设计一个基于高斯 (Gauss) 过程的过滤器从嘈杂的信息中提取最有价值的信息内容; 文献 [433] 在神经网络层面上研究多智能体通信系统的稳健性; 文献 [49] 将多智能体通信建模为一个双人零和博弈问题, 并应用 PSRO 技术来学习一个稳健的通信策略. ARTS [434], RADAR [435] 等工作考虑研究多智能体系统的弹性恢复能力, 研究协作多智能体强化学习任务在面对环境变化下的恢复能力.

最近一段时间, 针对环境动荡下协作稳健难的问题, 研究者通过对抗训练提升协作系统在测试环境发生改变时的稳健能力. 文献 [354] 提出演化对抗攻击者辅助的稳健协作算法 ROMANCE; 文献 [162] 提出通过种群对抗训练的对抗稳健多智能体通信训练框架 MA3C; 文献 [164] 提出多视图信息可验证性的稳健通信方法 CroMAC; 文献 [353] 针对多智能体观测噪声扰动进行研究; 文献 [436] 考虑针对多智能体中的状态攻击学习一个稳健策略.

4.3.2　多目标 (约束) 协作多智能体强化学习

多目标优化 [437] 是指在优化问题中存在多个目标函数, 需要同时考虑各个目标的最优解. 在多目标优化问题中, 不同目标函数之间可能存在相互矛盾的情况, 即改善其中一个目标函数可能会导致另一个目标函数的恶化. 因此, 需要在不同目标函数之间进行权衡, 找到平衡点或者称之为帕累托 (Pareto) 最优解集. 而在多目标强化学习中, 类似的, 智能体需要根据目标重要性进行权衡, 以学习得到对应的帕累托最优策略 [438~441]. 在多智能体强化学习中, 也有部分工作将多目标学习问题引入进来, 一般建模为多目标多智能体系统 (multiobjective multi-agent systems, MOMAS) [442], 其中不同目标下的奖赏函数可能相互冲突. 文献 [443] 考虑多智能体系统中的个体偏好与共享目标之间的关系, 将其建模多目标问题, 结果表明混合处理方式比单纯考虑单个目标取得更优效果. 文献 [444] 探讨了沟通和承诺如何帮助多智能体在具有挑战性的环境中学习到合适的策略. 文献 [445] 考虑一般博弈问题中的基于多目标的对手建模问题, 通过多目标加速策略学习.

另一方面, 最近部分工作关注于单智能体安全强化学习 [446, 447] 与多智能体安全强化学习 [257], 这类多约束问题在强化学习设定中可建模为受限马尔科夫决策过程 (constrained MDP, CMDP). 在多智能体强化学习中, 文献 [257] 提出针对多智能体任务的测试环境 Safe MAMuJoCo, Safe MARobosuite 与 Safe MAIG, 并针对提出安全多智能体强化学习算法 MACPO 与 MAPPO-Lagrangian. 文献 [448] 研究了在受限制的马尔可夫博弈中进行在线安全的多智能体强化学习, 其中智能体通过在对期望总奖赏进行约束的情况下最大化其期望总效用来进行竞争. 文献 [449] 研究了安全的多智能体强化学习, 其中智能体试图在满足自身安全约束的同时, 共同最大化局部目标的总和. CAMA [355] 研究多智能体协作中的安全问题. 文献 [450] 考虑状态存在扰动下的多智能体稳健与安全问题. 也有部分工作考虑基于屏障保护的安全多智能体强化学习 [187, 357, 358], 或将多智能体安全与控制技术相结合 [356, 451].

4.3.3　基于风险的多智能体协作

值分布强化学习 (distributional reinforcement learning, distributional RL) 近年来在多方面取得重要进展 [452]. 经典的基于价值的强化学习方法尝试使用期望值对累积回报进行建模, 表示为价值函数 $V(s)$ 或动作价值函数 $Q(s,a)$. 而在这个建模过程中, 完整的分布信息很大程度上被丢失了, 值分布强化学习为解决该问题, 对累积回报这一随机变量的分布 $Z(s,a)$ 进行建模. 该类方法在多智能体协作任务中也得到应用, 为了缓解由于局部可观测带来的环境的随机性, DFAC [359] 将单个智能体的收益函数由确定性变量扩展为随机变量, 然后将 QMIX 类算法的混合函数建模为配分混合函数, 该算法在多种高难度的协作任务上取得了较为优异的协作效果. 进一步, 为了缓解多智能体协作任务中的奖赏函数的随机性带来的不确定性, RMIX [360] 借助基于风险的值分布技巧 (conditional value at risk, CVaR) 提升算法的协作能力, 该算法创新性地提出基于智能体轨迹相似度评估的风险度评估, 理论证明 RMIX

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算法的合理性, 实验结果也进一步验证算法的有效性. ROE [153] 从另外的角度提出一种基于风险评估的协作动作乐观探索方法, 该方法可以针对性地对分布进行采样, 可以有效提升多智能体的探索效率.

除了上述所涉及的基于值分布的多智能体协作算法之外, 也有一些工作从其他方面展开, 比如基于值分布的奖赏评估 [361]、一般博弈问题中的自适应高效多智能体策略学习 [363]、多智能体学习过程中的风险解耦学习 [362]、基于博弈论的风险管理 [364] 等. 虽然所述工作在多种环境上取得了一定成果, 但是考虑到真实环境的风险未知, 如何将多智能体策略部署到真实环境中, 通过自动识别环境风险并相应调整协作策略是未来值得研究的方向之一.

4.4　智能体系统开放性

受开放环境影响, 多智能体系统中的个体成员可能会离开或突然加入到团队中. 不同的团队规模决定了系统的最佳协作模式, 而个体的不同风格需要智能体之间相互配合适应. 此外, 当要求与人类协作时, 智能体系统需要将人类偏好、行为风格、能力上限等因素纳入考虑范围内, 对策略提出了更高的要求.

4.4.1　自组织多智能体协作

自组织队友协作 (ad hoc teamwork, AHT) [50] 旨在创建一个自主智能体, 赋予其能够有效、稳健地与之前未知的队友进行高效协作的能力 [365]. 早期的工作假设需要配合的队友行为对于学习的智能体而言是已知的 [365,453]. 后续相关工作逐步放松该假设, 在交互过程中, 学习的自主智能体无法提前获知队友的行为. 一些工作设计算法通过观测队友的行为去预测相应队友的策略, 进而促进其协作过程 [454~457]. 另外一些工作试图通过有效的通信手段以进一步提升 AHT 中队友之间的协作能力 [458]. 以上工作虽然在一定程度上可以提升协作性能, 但是这些方法假设合作的队友处于封闭的环境中, 即队友在单轨迹内的数量和类型保持不变, 基于此, 开放 AHT 相应地被提出并研究 [332], 其中 GPL [335] 通过构建图神经网络应对不同时刻队友类型与数量的变化.

早期关于 AHT 的工作一般考虑 AHT 的智能体处于全局可观测的环境中, 最近的部分工作考虑将该设定扩展到局部可观测场景, 其中 ODITS [366] 设计基于互信息优化的正则项去评估其他队友的行为, 训练得到的自主智能体仅用其局部观测即可推测队友的行为. 与先前的研究不同, 文献 [459] 提出一种在部分可观测情况下解决开放式临时团队合作问题的方法. TEAMSTER [369] 提出一种解耦环境模型和队友行为模型学习的方法. 除此之外, 部分工作从其他方面展开工作, 包括存在攻击者的 AHT 问题 [460]、少样本交互协作 [52]、AHT 的队友生成覆盖问题 [367,368] 等.

4.4.2　零 (少) 样本协作

零样本协作 (zero-shot coordination, ZSC) 是近年来在协作多智能体任务中被提出来的概念, 其目标是训练智能体使其能够与未见队友协作 [51]. 自博弈 (或 “自我训练”) [461,462] 是有效提升协作能力的手段之一, 智能体通过与自身进行协作训练不断提升, 但是由此生成的智能体可能缺乏与未见队友协作的能力. 文献 [463] 进一步规范该问题, 添加顺序无关的训练方法以缓解次优性. 由于仅与单个队友训练配合可能存在过拟合到特定队友行为风格的问题, 其他方法如虚拟博弈 (fictitious co-play, FCP) [370,464], 或智能体和队友种群的协同演化 [372] 等技术都在该方面取得一定的成果. 也有部分工作利用少样本技术来应对多模态场景 [349,373] 并且取得一定的成效. 文献 [465] 最近通过合作队友的动作偏好对当前多种 ZSC 算法的容量进行评估与计算.

除了前述工作之外, ZSC 系列工作还包括多样性度量 [371,466]、训练范式设计 [370,372,463]、等量可交互网络设计 [467]、策略相似性评估下的协作提升 [468]、通用场景下的 ZSC 问题 [469]、基于集成技术提升的 ZSC [469]、人类价值偏好研究 [377]、多样队友生成 [374] 以及异构环境下的策略协同演化 [372] 等. 此外, 少样本适应在单智能体元强化学习中被广泛应用 [470~472], 在该设定下, 使用一个探索策略

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在环境中采样 K 条轨迹以获得对下游任务推断的隐变量 z 从而决策. 具体而言, 在少样本队友协作 (FST) [52] 中, 在不同任务中训练生成的智能体需要组成新的团队, 在未知但相关的任务中, 能够适应并协作完成任务. 其中 CSP [373] 考虑多模态协作范式, 开发一种协作策略与探索策略解耦的少样本协作范式, 在策略执行时采集少量样本以寻找最优策略头. 文献 [375] 发现当前性能较优的 ZSC 算法在面对不同的学习方法时需要较多的样本去适应新的队友, 相应提出一种基于少样本的协作方法, 并在 Hanabi 上验证算法的有效性. Macop [408] 考虑回合间协作对象发生变化下的策略适应能力, 提出面向任意队友的高相容性协作算法, 显著提升协作算法的泛化能力.

4.4.3　人智协同

赋予智能体 (机器人) 与人类高效协作的能力是人工智能一直以来的目标之一 [473, 474]. 其中人智协同 [241] 可以为人智交互 (human-AI interaction, HAI) [475] 或人机交互 (human-robot interaction, HRI) [476] 提供支持, 人智协同的目的是使得人类参与者与智能体之间更好地协作以完成特定任务, 针对不同的人类参与者, 可以通过协作多智能体强化学习提升人智协同能力.

与前述提到的 ZSC 问题不同的是, 人智协同考虑的合作对象是人类参与者. 虽然曾经有研究在部分环境下发现可能不需要人类数据训练智能体模型就能与真实人类合作 [370], 但在人类行为的某些微妙特征对任务产生至关重要影响的场景下, 是无法脱离人类数据产生良好协作策略的. 对于待训练的智能体而言, 可以通过先验偏置去直接编码具有人类行为风格的队友 [463, 477, 478], 也可以不同程度地利用与真实人类交互的数据进行训练. 一些方法则同时结合基于先验偏置手工编码人类行为与人类交互数据对智能体进行优化 [479~481].

然而, 上述方法都对测试时人类行为的模式做出较强假设, 在实际上难以成立. 针对这一问题, 涌现出一系列学习人类行为模型和计算对其的最佳响应的方法来实现人工智能与人类合作的方法 [376]. 在人智协同中, 目前有部分方法从其他角度进行研究, 包括研究人类有偏好的任务场景 [377]、通过离线数据促进人机协作 [378]、开发技术以实现人智相互配合 [379]、人机协作中的领导与跟随技术开发 [482]、零样本人智协作 [483]、基于贝叶斯优化的人智协作等 [484] 以及人机协作环境设置 [485] 等. 另一方面, 随着以 ChatGPT 为代表的大语言模型的发展, 目前也有部分工作尝试通过语言模型促进人智协作, 如 ProAgent [486] 提出一种可以充分利用大语言模型以进行队友行为预测的高效人机协作框架并取得优秀的协作性能. 虽然以上工作在人智协同方面取得一些成果, 然而该方向目前依然存在不少问题与挑战, 比如, 缺乏便捷且有效的测试环境, 大部分工作主要在 Overcooked [241] 上开展, 该测试环境存在智能体数量较少且部分场景过于简单的问题; 另外一些工作主要在第三方不开源的自制环境比如机械臂上进行验证, 如何在未来开发具备多种任务需求且适用人类参与者上手, 便于测试的测试环境, 以及设计更高效的算法等都是未来值得研究的方向. 另一方面, 通过人类价值对齐 [487, 488]、人在环训练 [489] 等, 是解决以上问题的可能方案之一.

5　总结与展望

本文着眼于协作多智能体强化学习的发展与研究, 从当前主流的经典环境到与现实应用贴近的开放环境下的多智能体协作逐层深入. 对强化学习、多智能体系统、多智能体强化学习、协作多智能体强化学习等方面进行概括性的介绍, 针对不同研究方向进行总结, 归纳凝练出经典环境下的多智能体强化学习研究关注点. 虽然目前许多基于封闭假设的多智能体强化学习算法在经典环境中取得了优异的协作效果, 但是在真实世界的应用仍然有限, 这很大程度应该归结于现有方法未对开放环境的特点进行针对性的研究, 离真正赋能日常生活还存在较大差距. 为克服开放环境下交互繁杂、系统多变、约束众多等障碍, 未来的协作多智能体强化学习可以从以下角度开展相关研究, 并基于此引发更多对开放环境下多智能体强化学习的关注, 让协作多智能体系统更好地被应用在现实环境中, 以赋能人类

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生活.

• **经典封闭环境下的多智能体协作问题解决.** 在经典环境下的协作多智能体强化学习算法是走向开放环境的基石, 通过提升系统在封闭环境下的协作性能, 可以让其具备更广的应用潜力. 然而目前封闭环境下的协作多智能体强化学习仍然存在诸多问题有待解决. 包括在智能体数量极多的大规模场景下进行高效的策略优化 [490], 如何平衡训练与执行过程中分布式与集中式的关系等, 都是未来研究工作需要考虑与解决的.

• **开放环境下的理论分析与体系构建.** 相较于封闭环境, 开放环境中的条件更严苛, 挑战更艰巨. 目前部分工作采用启发式规则对机器学习环境开放程度进行设计 [38]. 但是针对协作多智能体系统所处环境, 如何搭建包括多智能体协作开放性的完备定义、环境开放性程度定义, 以及算法的性能边界等在内的一套体系, 是未来需要进一步考虑的问题.

• **开放环境下的协作多智能体测试环境搭建.** 虽然当前有部分工作从稳健性等方面对开放环境下的协作多智能体强化学习展开研究, 但基准测试仍然是在经典封闭环境上进行修改所进行的, 仅考虑文章的特有设定, 缺乏兼容不同开放性挑战的测试环境与统一的评估标准. 如何搭建包含对前述 10 个方面进行评估的测试环境, 是未来促进开放环境下的协作多智能体研究的极大助力.

• **面向开放环境下的多智能体通用决策大模型.** 大模型, 尤其是大语言模型 [409] 近年来在多个领域引发广泛关注并且得到应用. 部分工作探索通用决策大模型 [411, 491] 并且在一些领域得到应用. 然而由于复杂繁杂、实体多变等原因, 多智能体通用决策大模型这一研究领域仍然存在较大空白, 在未来, 研究者需要考虑如何在多种任务场景下学习通用的多智能体决策大模型, 以实现零样本或少样本泛化, 或通过微调实现对未知任务领域的快速适应.

• **协作多智能体强化学习应用与落地.** 经典环境中多智能体强化学习的高效协作性能显现出极大的应用潜力, 然而大部分工作仅限于在模拟器或特定任务场景中进行测试 [492], 与真实的社会应用场景与需求目前还有一段距离. 而针对开放环境下的协作多智能体强化学习的研究的主要目的仍然是促进算法应用到人类生活并促进社会进步. 在未来, 如何将多智能体强化学习算法安全高效地应用在诸如大规模无人驾驶、智慧城市、大规模计算资源调度等领域, 都是十分值得探讨的话题.

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Progress on cooperative multi-agent reinforcement learning in open environment

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Abstract Multi-agent reinforcement learning (MARL) has gained significant attention and achieved significant progress across various domains in recent years. Within this field, cooperative MARL focuses on training teams of agents to collaboratively complete tasks that exceed the capabilities of individual agents. This approach has demonstrated immense potential in applications such as pathfinding, autonomous driving, active voltage control, and dynamic algorithm configuration. One of the key research areas in cooperative MARL is enhancing coordination efficiency in agents. Traditionally, most methods have been designed for simple, static, and closed environments. With advancements in artificial intelligence technology, emerging studies are beginning to explore multi-agent coordination in open environments, where the coordination settings for agents can dynamically change. These studies investigate scenarios involving varying environmental conditions, agent participation, and task complexities. While progress has been made, a comprehensive review and synthesis of research in this area remains necessary. This study begins by defining reinforcement learning, introducing MARL, cooperative MARL, typical methodologies, and testing environments. It then examines the evolution of multi-agent cooperation from closed to open settings, categorizes existing research, and highlights representative studies within this domain. Finally, it evaluates the strengths and limitations of this study and outlines future development opportunities and research directions of cooperative MARL in open environments. This review aims to inspire further engagement and discussions in this emerging and impactful field.

Keywords reinforcement learning, multi-agent system, multi-agent coordination, open environment machine learning, open environment multi-agent coordination

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